

Design Challenges in Silicon Photonics Mach-Zehnder Modulators

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Design Challenges in Silicon Photonics Mach-Zehnder Modulators

by

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Abstract

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Communication plays a vital role in people's life. As more people are relying on computing power and artificial intelligence system to finish their daily tasks, high-speed communication methods are required. Also, as the current crisis of COVID-19 spread, people had to work and study remotely to stop the spreading of the virus. All this increased reliance on telecommunication systems requires high bandwidth to support millions staying at home. This highly interconnected world and the easiness of communication using the internet have transformed the way people communicate and lead to the creation of new social platforms such as Facebook, Instagram, and Snapchat. Due to the intense usage of those applications and the internet in general, a tremendous amount of information from people interacting with the internet is gathered. The increasing amount of data gathered per day lead to the creation of the data mining field. Data mining is the process of searching, filtering, analyzing, patterns in large data sets in order to understand and discover patterns by using an Artificial intelligence (AI) algorithm to identify the patterns.

Communication is mainly optical communication, which is the backbone of the internet. On a device level, an Optical Modulator (OM) is used for on-chip communication. It is used to modulate a beam of light that can be carried over free space or propagating through an optical waveguide. As computing power increases, it is required to have a reliable communication method to communicate and send data between different parts of the chip itself. There are different optical modulator designs, most commonly based on the Mach-Zehnder

Interferometer (MZI). These modulators are therefore called Mach-Zehnder Modulators (MZM). Having a reliable OM over a large range of temperature while making it as small as possible is a challenge due to the high thermo-optical coefficient (TOC) of Silicon around $1.86 * 10^{-4}K^{-1}$. Due to the high TOC of Silicon, the optical modulator can operate on a narrow range of temperatures. The work developed toward this thesis is aimed at designing an athermal optical modulator with ultra-high-speed modulation. An athermal optical modulator is an optical modulator where the modulated signal will not be affected if the device operates under the designed range of temperature. Two designs are proposed and simulated in this thesis. The first design is an athermal Mach- Zehnder Modulator, which is achieved by carefully choosing different waveguide widths and arm lengths that lead to an almost zero change of optical path difference with temperature variations. The second design is a Ring Resonator (RR) assisted Mach-Zehnder Interferometer (MZI). The lengths of the arms and the radius of the RR are chosen carefully, so the RR will passively compensate for the optical path change of MZI with temperature. Different doping profiles were studied to achieve the highest modulation speed while keeping the device thermally stable during the operation.

Indexing Terms: Mach-Zehnder Modulator, Ultra-high-speed modulator, Ring Resonator, Silicon Photonics

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Declaration and Copyright

Declaration

I declare that the work in this thesis was carried out in accordance with the regulations of Khalifa University of Science and Technology. The work is entirely my own except where indicated by special reference in the text. Any views expressed in the thesis are those of the author and in no way represent those of Khalifa University of Science and Technology. No part of the thesis has been presented to any other university for any degree.

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Nomenclature

CMOS	Complementary metal-oxide-semiconductor
IC	Integrated circuits
PIC	Photonic Integrated Circuit
SOI	Silicon on insulator
TE	Transverse electrical
TM	Transverse magnetic
MZI	Mach-Zehnder interferometer
MZM	Mach-Zehnder modulator
RR	Ring resonator
GDS	Graphic data stream
GF	GlobalFoundries
QCSE	Quantum confined Stark effect
EAM	Electro absorption modulation
RRM	Ring resonator modulator
FWHM	Full width half maximum
PAM	Pulse amplitude modulation
NRZ	Non-return to zero
TOC	Thermo-optical coefficient
OP	Optical path
FSR	Free spectrum range
ER	Extinction ratio
SRIM	Stopping and Range of Ions in Matter

Chapter 1: Introduction

1.1 Motivation

In the 21st century, internet connection is as essential as electrical power in our daily lives to run computers and phones. Internet connection plays a crucial role in our lives in different areas such as:

- Communications: As the internet connects people around the world, it helps in instant communication. People do not need to wait for a long time for their letters to reach their destination as it used to be.
- Education: Many people today use the internet for distance online education where they can get an academic degree while they sit at home.
- Research: Students use the internet to find papers, journals, and different publications on the topic of interest.
- Work and Finance: Working environment has been affected a lot by the internet as people can work from a distance and get paid without even leaving the house.

Moreover, there are many more areas that the internet has changed in the past years. Also, it created a highly interconnected world.

This highly interconnected world and the easiness of communication using the internet have transformed the way people communicate and lead to the creation of new social platforms such as Facebook, Instagram, and Snapchat. Due to the intense usage of those applications and the internet in general, a tremendous amount of information from people interacting with the

internet is gathered. The increasing amount of data gathered per day lead to the creation of the data mining field. Data mining is the process of searching, filtering, analyzing, patterns in large data sets in order to understand and discover patterns by using an Artificial intelligence (AI) algorithm to identify the patterns.

Optics has a vital role in internet communication, as fiber optics is the backbone of the internet. It is the method of transmitting information from one place to another by sending light pulses through an optical fiber. While optical fibers transmit the data over long distances, for example, transmitting the data from one continental to another. The optical modulator modulates and transmits data on a chip level by propagating through an optical waveguide by manipulating a property of light, such as its intensity, polarization, or phase. This manipulation of the light property will be used to transmit data.

1.2 Silicon Photonics

The microelectronic industry continues to grow as Moore's Law observed that the number of transistors doubles every 18 months [1]. This observation leads to an increase in processing power speed by decreasing the size of the transistor, which is a challenge, and due to that, Moore's Law has declined in the past years. Silicon integrated devices have the potential to deliver the functionality required by future devices [1]. Silicon has been the primary material used in electrical circuits, and due to its advantages in optics, integrating the electrical and optical world on a chip is a promising new technology that will increase the data transfer speed.

As the size of the transistor decreases, the processing speed increases, however the same is not valid for the electrical connections; by reducing the electrical copper wires size, the cross-talking increases alongside the RC time constant and power dissipation. Cross talk increase due to the coupling with neighboring wiring at the same time, the resistance increases; therefore, the RC time constant increase.

On the other hand, the optical connection removes most of the problems associated with electrical wiring. Light (photons) travels in a waveguide or optical fiber without being affected by RC constant and with a minimum amount of heat dissipation. Therefore optical communication generated great interest to search and advance in using it not only on a large scale to communicate from continent to another (as the case of current optical fibers used to connect the internet around the world) but also on a small scale where they can be integrated with electrical circuits used as communications method for different circuits or devices.

Silicon photonics technology has incredible progress in the past few years, with several groups designing microscale devices the perform the optical functions of centimeter-sized table-top optics and millimeter-sized fiber optics components. These efforts are motivated by the high bandwidth and high speed that can be achieved compared to traditional electronic devices and the enormous possibility of optoelectronic integration. Silicon is the basic material used in Complementary Metal-Oxide-Semiconductor (CMOS) process industry with mature and low-cost technologies [2]. Silicon has a high refractive index (ratio of the speed of light in a vacuum to the speed of light in a material), and by surrounding it with low refractive index material such as silicon dioxide which is an excellent isolator which is silicon's native oxide, the combination of silicon and its oxide creates an excellent confined optical mode. Also,

silicon has an indirect bandgap, and several nonlinear effects tune its refractive index. Some of the nonlinear effects are:

- Free-carrier absorption and dispersion effects [3], [4];
- Thermo-optic effect [5];
- Spontaneous and stimulated Raman scattering effects [6], [7].

In this thesis, both the free-carrier absorption and dispersion effects and the Thermo-optic effect will be studied alongside their effect in tuning the refractive index is enormous. Also, those effects directly related to the design proposed in the thesis.

By manipulating the nonlinear effects of silicon, many devices were built based on this platform, including optical amplifiers [8]–[10], optical modulators [11]–[15], logic gates [16], [17], optical couplers and optical detectors. While some of the mentioned devices use only silicon and its native oxide, other devices are developed by silicon or silicon compatible materials. By using different devices mentioned above, exciting applications have been presented, such as routers [18], [19], transmitters, and receivers [20], [21].

1.2.1 History of Silicon Photonics

Silicon is available in large quantities around the world as it is the second most abundant element of Earth. With atoms arranged in cubic crystal structure, a wafer can be made from it with tremendous purity with almost zero defects by either Czochralski or float-zone crystal growth methods. As mentioned in the previous section, silicon has a high refractive index of about 3.503 at 1310 nm wavelength and is transparent to low energy infrared light. Due to its high index, devices made with silicon can be downscaled to the order of micrometer and

nanometer sizes. Also, due to its native oxide (SiO_2) with its low refractive index of about 1.4468 at 1310nm offers an excellent cladding to the silicon in order to create a waveguide that can be used for the optical signal to travel in [22].

Jack Kilby at Texas Instruments demonstrated the first (hybrid) integrated circuit (IC) in 1958. The next year, Robert Noyce at Fairchild Semiconductor invented and demonstrated the first monolithic integrated circuit. These circuits can be found in almost every present-day electronics appliance. High quality, the inexpensive silicon wafer, is the core of the IC industry. In the late 1980s and in mid-1990s, a new type of wafer was created known as Silicon on Insulator (SOI). SOI substrates are the perfect choice for creating waveguides due to the ability of highly confining the light in the top layer.

Following the leading work presented by Soref et al.[24], [25], early stages of research on silicon photonics focused on passive and active waveguides, passive waveguides as a means to transmit the light from one end to another, while active photonics such as modulators, which is the core topic of this thesis, was not presented until 2004 [22]. Figure 1.1 shows the

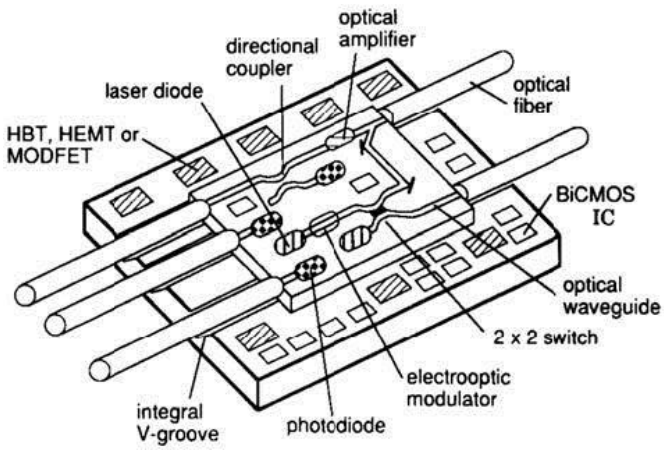


Figure 1.1: Silicon-based optoelectronics integrated circuit (OEIC) Superchips presented by Soref *[23]*.

Superchip work presented by Soref for creating a silicon-based optoelectronics integrated circuit [23].

1.2.2 Passive devices

Passive devices are critical components of silicon PICs (Photonic Integrated Circuits) where no energy is required to fulfill their operation. Different devices have been researched to be used as passive devices such as waveguides, couplers, Bragg gratings, Mach-Zehnder interferometers, ring resonators, and more. Some of the passive devices will be discussed in the following sections.

1.2.2.1 Waveguides

In the 1980s, the first waveguide on silicon was reported by Soref [24], [25]. Waveguides are the basic building block of silicon photonics. It is created by having a high index material such as silicon, forming a core, surrounded by low index material such as silicon dioxide, which is called cladding. Waveguides are responsible for carrying the signal as the light is confined by the difference of the refractive index of the core and the surrounding material (cladding). As the SOI platform was introduced, waveguide applications have adopted mature silicon microelectronics fabrication technologies. The first micrometer-thick SOI slab waveguide was proposed in 1988 [26]. The first single-mode rib waveguide in SOI with the width and height of several micrometers was reported in 1991 [27].

As the size of the waveguide was reduced and reached the nanoscale, another issue was essential to be addressed, which is the scattering loss. As light propagation was profoundly

affected by scattering, research has been conducted to improve the waveguide design and fabrication and reduce the scattering loss. Some papers suggested the use of oxidation to reduce the sidewall roughness and the corresponding loss [28]. Another method has been used to define silicon waveguides with various oxidation processes [29]–[31].

The high contrast of the SOI platform offers its advantage in a highly confined light mode in micro/nanoscale waveguide dimensions. However, a drawback is introduced, and it is representing a challenge for commercial applications. The drawback is called waveguide birefringence, which is the difference of effective index between the Transverse-Electrical (TE) and transverse-Magnetic (TM) polarization [22]. This waveguide birefringence has significant drawbacks, as both polarization propagates with very different phase and energy velocities, leading to polarization-dependent dispersion and loss and also affect the performance of many interferometric based devices, which rely on the phase-matching condition, which can only be met for one polarization at a time.

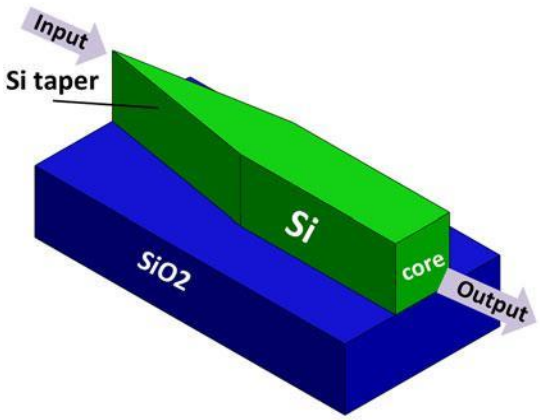


Figure 1.2: Illustration of an inverted taper coupler.

1.2.2.2 Couplers

Fiber-to-waveguide coupling is a critical issue as there is a significant mode mismatch between them. More than one solution has been presented to solve this issue. One of the solutions is to have an inverted tapper [32], [33], as shown in Figure 1.2. Another solution is by using a grating coupler to couple the light between the fiber and the waveguide. A grating coupler is a periodic feature that assists the light to couple to the waveguide in near-normal orientation [22]. This method minimizes the back-reflection of the light, and the well suited for wafer-level probing, without the need for chip dicing and complex 3D alignment procedures.

1.2.2.3 Polarization splitters/rotator

The main use of the polarization splitters/rotator is to eliminate the disadvantage of having TE/TM mode. As each mode propagates with a different index, splitters are used to split TE from TM, while rotators are used to change one polarization mode into another. The types of basic polarization splitting, and rotation principles are:

- Mode-coupling [34].
- Mode-evolution [35]–[37].

1.2.3 Active devices

1.2.3.1 Modulators and switches

Modulator and switches are one of the primary active devices on silicon. As a modulator can manipulate properties of transmitted light, it can modulate an electrical signal

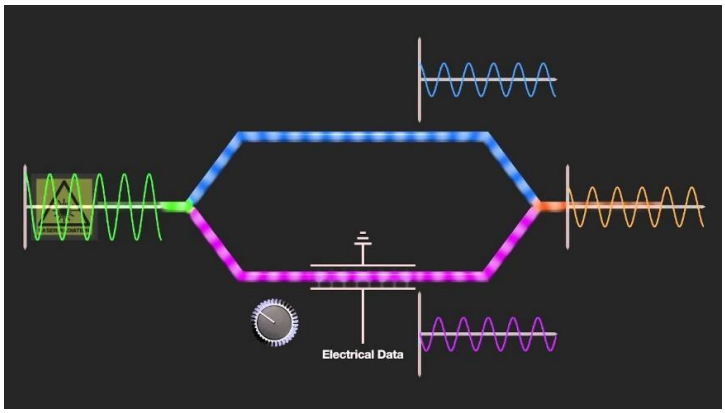


Figure 1.3: Mach-Zehnder modulator illustration

into an optical signal. On the other hand, an optical switch can selectively switch to route the optical signal to one or another circuit. Different kinds of modulators can be categorized in:

- Amplitude modulators
- Phase modulators
- Polarization modulators

Typically, the modulator encodes a signal onto the light propagating through it by changing the intensity of the light at the output, using the power supplied by an electrical signal. Figure 1.3 illustrates the design of the Mach-Zehnder modulator (MZM).

1.2.3.2 Photodetectors

Photodetector usually placed at the end of the optical interconnect in the circuit as it converts an optical signal back to an electrical signal. A photodetector absorbs the photon received and generate free electron-hole pairs. The output of the photodetector is an electrical current. This process happens when the energy of the incident photon received is equal or higher than the bandgap of the material used in the photodetector. There are different

parameters to be considered in photodetector such as quantum efficiency which relate the number of incident photons to the number of electron-hole pairs generated [22], responsivity which is the ratio of the output current to the light power reaching the photodetector, dark current which is the current generated by the photodetector even in the absence of light hitting the detector and material compatibility for integration in the fabrication process of interest.

1.3 Dissertation Organization

The organization of this thesis is as follows. Chapter 2 is a detailed background about silicon optical modulators, where modulation schemes and limitation of electrical connections are discussed. The advantages of optical modulation over electrical alongside with the challenges of the optical modulation is reviewed. Also, the bandwidth of the current systems is discussed alongside with the optical system for 400 Gbits/s and above. The second part of this chapter is a detailed review of different types of modulators: MZI based modulators, Ring Resonator (RR) modulators, and MZI – RR modulators. Finally, the chapter describes the design challenges faced when designing the modulator.

In Chapter 3, Athermal MZI design: the theoretical description of athermal operation is explained by defining all the fundamental properties of the theory, such as effective index, group index, and thermo-optical coefficient). An example design of MZI in GF-45SPCLO technology is proposed and another design in GF-90wg technology. The differences between the designs in each platform will be discussed.

In Chapters 4, The modulator design is discussed with the doping profiles and their modeling. Also, the free carrier dispersion effect described by Soref [3], [4] is discussed and

compared (papers of 1987 and 2011). A charge distribution is modeled, and the effects of the charges are simulated using an FEM charge solver alongside with Mode solver to observe the effect of the doping profiles on the effective index of the simulated structure. Voltage characteristics and optical performance are simulated and observed to redesign the device if needed to achieve the required goals of the design. The optical interconnect model is modeled and simulated to observe the expected optical performance. Finally, Graphic Data Stream layout (GDS) is created and modified for tape-out by Global Foundries (GF).

In Chapter 5, the thesis is summarized, concluded, and future work is presented.

Chapter 2: Background and Review

2.1 Limitations of electrical Interconnects, importance, and advantages of optical interconnects

Computational speed is demanded to be increased as the world is generating a massive amount of data by interacting with the internet daily. This demand will increase the number of cores on a chip, and the data transfer between cores will exceed 1 Tb/s [38], [39]. Over the years, electrical/electronic interconnects have shown some improvements in different areas such as bandwidth, power consumption, and latency. However, as the size of transistors is getting smaller, the device area is fundamentally limiting the maximum bandwidth. Also, as the transistors get smaller, the electrical copper wire suffers from an increased crosstalk effect. Furthermore, the power consumption of the interconnect increases and expected to rise to >80% of the total chip power, which is limited to the total power of 200W [40]. To summarize, the main limitation of the electrical interconnects is dispersion, crosstalk, speed, bandwidth, and power consumption.

The success in silicon photonics, especially in modulator is essential due to the massive number of applications that needs a reliable high bandwidth device. Those applications include biological, chemical, medical, military, and astronomy applications [41]. Also, optical interconnects offer many advantages over electrical counterparts in different areas such as by using optical interconnects intricate designs can be achieved by introducing enabling modules and optimized cooling. Energy-saving is an essential factor as electrical interconnects use more

power compared to optical interconnects. Also, using optical interconnects precise clock and signal timing can be achieved [42].

2.2 Modulation schemes

Modulation is the process of manipulating one or more properties of a waveform called carrier wave/signal, which is typically a sinusoidal wave and mixing it with another signal that holds the data needed to be modulated known as data/message signal. In other words, mixing a low-frequency signal with a high-frequency carrier signal is called modulation. The equation below represents a sinusoidal wave:

$$f(T) = A\ sin(\omega t + \phi)$$

Where:

- A = the amplitude of the wave.
- ω = Frequency.
- ϕ = Phase angle.

The main difference between the carrier signal and the data/message signal is that the carrier signal does not hold data on its own and, in some cases, is not suitable to be transmitted. On the other hand, the carrier signal can be transmitted easily, and that is the main reason behind modulation. The basic types of modulation are explained in the following Table 2.1.

Table 2.1: Comparing different types of modulations

Type of modulation	A (wave amplitude)	ω (wave frequency)	ϕ (wave phase angle)
Amplitude	Modulated	Constant	Constant
Frequency	Constant	Modulated	Constant
Phase	Constant	Constant	Modulated

As described in Table 2.1 above, the main difference between the type of modulation is which property of the carrier signal is used to modulate (carry) the data signal.

Modulation can be categorized into analog modulation and digital modulation. As an example, for analog modulation is the audio signal modulated over an analog bandpass channel, while digital modulation is modulating data streams over an analog channel. As this thesis is focusing on optical communication (digital data streams to be communicated using optical medium), a brief description of different schemes of digital modulation will be discussed.

On-Off Keying (OOK) is the most basic form of amplitude modulation as data cannot be sent in terms of one's and zero's (On and Off). Sending a bit of data representing one, the modulated signal is on, while to send a zero bit of data, the modulated signal is off (No carrier is present at this state). This modulation is commonly used in Morse code. This modulation scheme is popular because of the simplicity and low cost of implementation, as in the off state, there is no transmission of any data. Thus, power consumption is lower than other modulation schemes.

Pulse-amplitude modulation (PAM) is an amplitude modulation scheme where each bit (zero or one) modulated to a corresponding level in an analog signal. Theoretically, the number

of possible PAM configurations is infinite. For example, PAM-4 is a 4-level pulse modulation. That means it possible to modulate two different signals using PAM-4. Generally, it can be said that PAM-(n), where n is the level of modulation follows the equation:

$$n = 2^x$$

Where x is the number of modulated signals. In other words, in PAM-8, there are 8-levels of modulation equal to 2^3. Thus, three different signals can be modulated using this modulation scheme.

Figure 2.1 shows how PAM-4 modulation works. By modulating two signals, a 4-level modulation is obtained (every two bits are mapped to a level). Table 2.2 shows an example of two signals and how they correspond to each level in PAM-4 modulation.

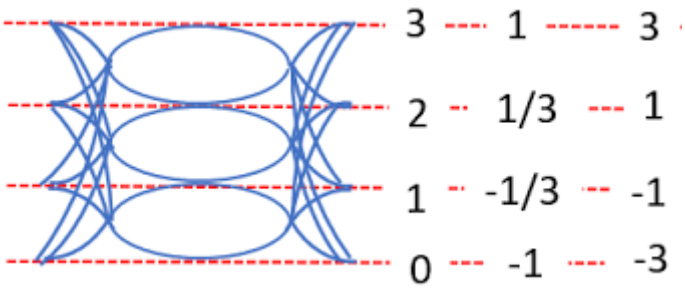


Figure 2.1: PAM-4 Modulation illustration.

Table 2.2: PAM-4 Modulation truth table.

First Signal data	Second Signal data	PAM-4 Level
0	0	0
0	1	1
1	0	2
1	1	3

2.3 The bandwidth of current systems

The maximum rate of data transfer is called bandwidth. This property is commonly used in communication systems, signal processing, wireless communication, and much more. Typically, bandwidth is measured in hertz, which is the range from the minimum and maximum frequency where the data is transmitted. Also, it is known as the channel capacity of the physical communication path, which is the waveguide in this thesis. Sometimes the bandwidth is defined as the total bits transferred per second in a channel.

The bandwidth used in devices can be divided into two categories narrowband and broadband (known as wideband). Figure 2.2 shows an illustration of the two bands. Applications desire a particular set of parameters to be achieved: high data rate, minimum spectrum usage, lower power consumption, most extended transmission range, and high resistance to noise. However, in order to increase the data rate, the frequency spectrum must be increased. This strategy will lower power consumption, range, and resistance to noise (interference). These tradeoffs are the main difference between a narrowband, which uses a small range of frequency compared to broadband. Table 2.3 shows a simple comparison between narrowband and broadband.

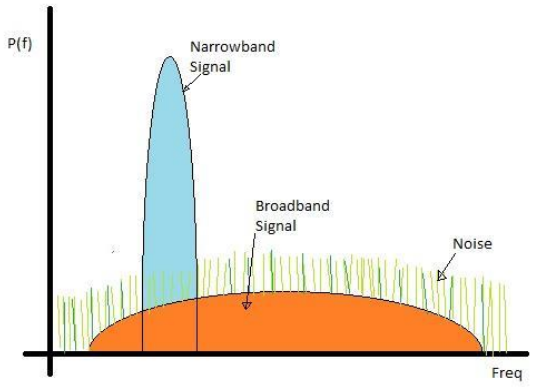


Figure 2.2: Narrowband and Broadband spectrum illustration.

Table 2.3: Narrowband and Broadband comparison.

Parameter	Narrowband	Broadband
Spectrum efficiency	Very High	Low
Power efficiency	High	Low
Data rate	Low	High
Transmission range	Very Long	Less

Table 2.4 and

Table 2.5 compare between different technologies and their bandwidth alongside with when this technology was invented for narrowband and broadband.

Table 2.4: Narrowband technology data rate and year [43].

Technology	Rate (Kbit/s)	Year
Morse code (used by a skilled operator)	0.021	1844
Modem 110 baud (Bell 101)	0.11	1959
Modem 300 (300 baud; Bell 103 or V.21)	0.3	1962
Modem 1200 (600 baud; Vadic VA3400, Bell 212A, or V.22)	1.2	1976
Modem 2400 (600 baud; V.22bis)	2.4	1984
Modem 9600 (2400 baud; V.32)	9.6	1984
Modem 14.4 (2400 baud; V.32bis)	14.4	1991
Modem 28.8 (3200 baud; V.34-1994)	28.8	1994
Modem 33.6 (3429 baud; V.34-1996/98)	33.6	1996

Modem 56k (8000/3429 baud; V.90)	56.0	1998
IDSL (dual ISDN + 16 kbit/s data channels)	144	2000

Table 2.5: Broadband technology data rate and year [43].

Technology	Rate	Year
ADSL (G.lite)	1536 Kbit/s	1998
SHDSL ITU G.991.2	5690 Kbit/s	2001
ADSL (G.dmt) ITU G.992.1	8192 Kbit/s	1999
ADSL2 ITU G.992.3	12.288 Mbit/s	2002
ADSL2+ ITU G.992.5	24.576 Mbit/s	2003
DOCSIS 2.0 (cable modem)	38 Mbit/s	2002
VDSL ITU G.993.1	52 Mbit/s	2001

VDSL2 ITU G.993.2	100 Mbit/s	2006
Uni-DSL	200 Mbit/s	2006
EPON (802.3ah) fiber optic service	1000 Mbit/s	2008
GPON (G.984) fiber optic service	2488 Mbit/s	2008
DOCSIS 3.1 (cable modem)	10 Gbit/s	2013
10G-PON (G.987) fiber-optic service	10 Gbit/s	2012
NG-PON2 (G.989) fiber-optic service	40 Gbit/s	2015

2.4 Physics of Silicon Optical Modulators

The optical modulator is a device that modulates light by changing some of the properties of the light to produce a modulated signal. The light can be transmitted through free space or in a waveguide. The optical modulators can be categorized into three main categories: amplitude, phase, or polarization modulators [41]. Also, modulators can have different types depending on the nature of operation they use: electrorefractive and electroabsorptive. To understand how the optical modulator works, it is required to understand different properties of silicon, which will be described below in this section.

2.4.1 Refractive index

The refractive index is the main property that affects the light and how it travels in the material. By manipulating this property, a modulator works as required. The Refractive index, which is also known as the index of refraction, is a dimensionless number used to describe how

fast the light is traveling through the material. In other words, it is the ratio of the speed of light in the vacuum to the speed of light in material for a specific wavelength λ.

$$n = \frac{c}{v}$$

The above equation is used to calculate the refractive index of a specific material, where n is the refractive index of the material, and c is the speed of light vacuum, and v is the speed of light in the material. For example, for a wavelength of 1310 nm (O – band), the refractive index of silicon is approximately 3.5, which means that the light travels 3.5 times slower in silicon than in vacuum at the mentioned wavelength.

Refractive index (n) is a complex number defined as $n = n + ik$. This appears due to the fact that most semiconductors have the permittivity function to be complex such as the equation below:

$$\varepsilon(\omega) = \varepsilon'(\omega) + i\varepsilon''(\omega)$$

The real part of the refractive index is the part explained above. While the imaginary part (k) is known as the extinction coefficient and its linked to the absorption coefficient (α) of the material by the following equation:

$$\alpha = \frac{4\pi k}{\lambda_0}$$

Where λ_0 is the free space wavelength of light. This relation shows that the extinction coefficient and the absorption coefficient is directly proportional [44].

2.4.2 Refractive index Change and attenuation change for silicon

In order to change the refractive index, the material should be doped with implants so that the free carrier concentration of the implants affects the refractive index and change it when a voltage is applied. A study of the effect of the free carrier on the refractive index of silicon is done by Soref [3], [4]. The details of the study will be explained thoroughly in section 4.1. At this stage, knowing that the voltage applied over a doped region of silicon can change the refractive index of the material is enough to understand how the modulator works. The following equations describe the change in refractive index at a wavelength of 1310 nm [4]:

$$\Delta\alpha = \Delta\alpha_e + \Delta\alpha_h = 2.48 * 10^{-22}\,\Delta N_e^{1.229} + 1.02 * 10^{-19}\Delta N_h^{1.089}$$

$$\Delta n = \Delta n_e + \Delta n_h = -2.98 * 10^{-22}\,\Delta N_e^{1.016} - 1.25 * 10^{-18}\Delta N_h^{0.835}$$

Where:

- α = the absorption coefficient.
- n = the refractive index.
- $\Delta\alpha_e$ = change in absorption due to change in free-electron carrier concentration.
- $\Delta\alpha_h$ = change in absorption due to change in free hole carrier concentration.
- Δn_e = change in refractive index due to change in free-electron carrier concentration.
- Δn_h = change in refractive index due to change in free hole carrier concentration.
- ΔN_e = change of free electron concentration in cm^{-3}.
- ΔN_h = change of free hole concentration in cm^{-3}.

The equation mentioned above is straightforward to calculate the change in the refractive index or the change in absorption in regard to the changer of the carrier concentration. In order to interact with the light propagating through a waveguide it requires an electrical manipulation of the charge density, which can be achieved by using different methods such as carrier injection, accumulation, or depletion. The main focus of this thesis is to change the carrier concertation by changing the depletion region. As the reverse voltage applied to the depletion region increases, the region increases, and the concertation is changing; thus, the refractive index of the material is changed.

To create an efficient modulator, its essential to achieve a significant change in the refractive index, thus a big change in the phase shift while keeping the absorption losses at a minimum. As it is evident from the above equations, as we increase the carrier concentration, the change of the refractive index is more prominent, but the absorption coefficient increases too. Comparing the effect from holes and electrons shows that it is preferred to create a waveguide predominantly P-type as the effect of holes will have a smaller impact on the absorption coefficient compared to electrons.

Another modulation mechanism in different material has been recently investigated, such as germanium. Those new studies aim to achieve a more efficient modulation. The Franz-Keldysh effect [45] and the quantum-confined Stark effect (QCSE) are the main studied effects, which are electric induced changes in optical absorption [41]. The Franz-Keldysh effect is weaker than QCSE, which has more abrupt and more substantial changes in the absorption coefficient [40]. However, in order to use those effects in silicon is required to add another semiconductor material to be used alongside the silicon, which complicates the design and structure. In 2005, Kuo et al. presented a demonstration of using QCSE with pure Ge quantum

wells and SiGe barriers on Silicon [46]. The proof-of-concept of electro-absorption modulation (EAM) was later on demonstrated using similar structures using the QSCE mechanism [47], [48].

2.5 Design of Silicon Modulator

In this section, the change of the refractive index of the material will be described. Changing the refractive index of the material can lead to a phase shift of the propagated light, which lead to intensity modulation. The state of the artwork will be discussed for different types of modulators and compared together. The main focus will be on MZI based modulators, RR based modulators, and MZI – RR assisted modulators.

2.5.1 Phase modulation

Free carrier distribution can be changed by electrical manipulation, and this is achieved in three ways for plasma dispersion silicon modulator: carrier accumulation (the device is forward biased), injection, and depletion (the device is reversed biased) [41]. Waveguides with a thin layer of insulation, typically SiO_2 acting as separation of the waveguide in the forward-biased mode (carrier accumulation) realized as a capacitance structure. This is due to the accumulation of electron and holes on either side of the insulating layer at the interface. This results in to change of carrier concentration, therefore the refractive index changes leading to phase change. Furthermore, removing the thin layer of oxide leads to injection design and change in carrier concentration in the doped waveguide. An advantage of the injecting design is the high modulation efficiency [49], but it is limited by the lifetime of the minority carriers

[50]. Another issue with the device being in forward biased mode is that injected carriers recombine with minority carriers during the carrier's flow, and this limits the switching speed of the modulator.

Depletion mode where the device is under reversed biased can eliminate the issue of the carrier lifetime. Also, this mode offers a high-speed performance [51].In the depletion mode, the waveguide is doped into p and n regions to create a PN junction, and the reverse voltage is applied to the junction. As the reverse voltage increases, the depletion region size increases. Thus, the carrier concentration changes, therefore the refractive index changes. According to Soref model, the change of the refractive index is more significant with the hole than the electrons [4]. The following equations describe the size of the depletion region for each of the following cases:

Case A: No voltage is applied, and the depletion region is purely due to the formation of PN junction [44], [52]:

$$x_p = \frac{N_D}{N_a+N_D}\, x_w \qquad\qquad x_n = \frac{N_a}{N_a+N_D}\, x_w$$

$$x_w = \left[\frac{2\varepsilon_p V_0}{qN_aN_D\left(N_D + \frac{\varepsilon_p}{\varepsilon_N}\, N_a\right)}\right]^{\frac{1}{2}} (N_a + N_D)$$

Where x_p is the depletion region in the p-type, x_n is the depletion region in the n-type, N_a is accepter doping concentration, N_D is donor doping concentration, ε_p and ε_n are the permittivity in p and N region respectively and V_o is the contact potential or known as the built-in potential voltage.

Case B: A voltage is applied, and the depletion region changes with the applied voltage.

$$x_w = \left[\frac{2\varepsilon_p(V_0 - V)}{qN_aN_D\left(N_D + \frac{\varepsilon_p}{\varepsilon_N}N_a\right)}\right]^{\frac{1}{2}}(N_a + N_D)$$

Where V is the applied voltage. As it is observed from the above equation, it is clear that if the voltage applied is <0 (reverse biased), the depletion region will get bigger, and that is what used in most on the modulator devices. Figure 2.3 shows an illustration of the depletion region.

The active length (L) required for the phase to have a phase shift of π can be described by:

$$L_\pi = \frac{\lambda}{2\Delta n}$$

Where Δn is the change in refractive index and λ the wavelength of operation [51]. In the depletion mode where the device is reversed biased, the phase shift increases with the increase of the voltage applied (reverse bias), but this relation is not indefinitely linear. Figure 2.4 shows the change in the depletion region width with voltage. The change of depletion width is

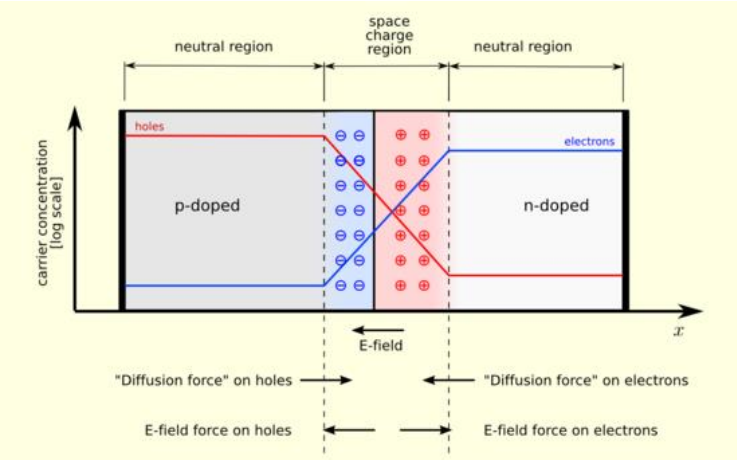


Figure 2.3: Depletion region schematic.

more significant at lower voltages than higher ones. In other words, the change of depletion

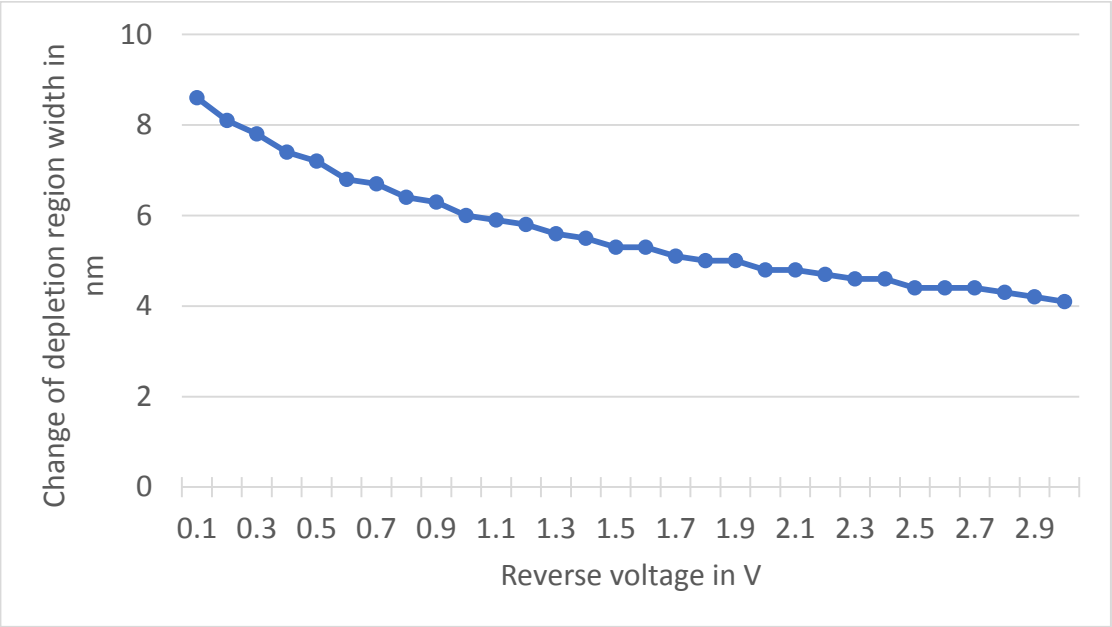


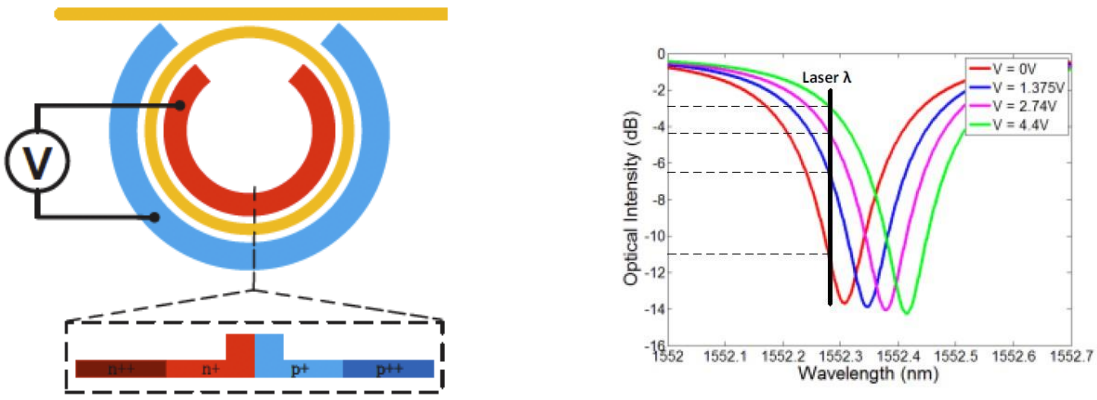
Figure 2.4: Change of the depletion region as a function of reverse voltage.

region width is smaller from -2 V to -3 V than it is from -1 V to -2 V. This means at higher voltages the refractive index cannot be changed; thus, the phase shift will not also change (saturation is reached). This shows that it is crucial to optimize the doping profile to ensure the optimal phase shift is achieved by maximizing the free carrier effect.

2.5.2 Intensity modulation

In the previous section, the phase shift achieved by changing applying a voltage in reverse bias was described. To translate this phase shift into intensity modulation, a structure is needed to accomplish the intensity modulation with is either an interferometric structure or a resonant structure. The most used devices are the ring resonator (RR) and Mach-Zehnder interferometer (MZI).

In a ring resonator, the modulation is achieved by altering the refractive index; thus, a change in resonance condition is obtained at the wavelength of operation known as the central wavelength. The advantage of the RR is the small footprint, but the optical bandwidth is limited as the resonance is operating on a specific wavelength [41]. Another disadvantage of using RR is it is sensitivity to temperature and fabrication errors, and the temperature sensitivity is due to the high thermo-optic coefficient of silicon [5], which will affect the refractive index in the opposite sense to the change produced by free carriers [51]. Figure 2.5a shows a schematic of the RR modulator, the doping is taking around 75% of the ring, and the voltage is applied across the doped region where the free carriers are presented. A cross-section of the waveguide is shown to observe how the doping regions n-type and p-type are presented. Figure 2.5b shows how the optical power in dB is transmitted for the central wavelength in the ring modulator as a function of voltage in the depletion mode.



(a) *(b)*

Figure 2.5: Ring Resonator modulator (a) illustration of RR modulator with the doped region and voltage connection [53] *(b) optical response of ring resonator as a function of voltage at a central wavelength* [53].

Mach-Zehnder modulator (MZI) is the most dominating and widely used type of intensity modulators [54]. Figure 2.6a shows a basic diagram of the MZI modulator, where the

input light is split into two branches by a Y-junction and send into two different waveguides. One of the arms is doped to use the free carrier effect to change the refractive index, thus change the phase of the wave traveling through it by applying a voltage across the doped region. Combining the two arms to using another Y-junction to allow the two waves to interfere and modulation occurs. The output transmission is shown in Figure 2.6b. When both the waves interfering is in phase (no phase difference between the waves), the optical power out is normalized to one, while if the waves were out of phase by π radians, the optical output power is zero. The following equation can describe the output of the MZI:

$$I_o = \frac{I_i}{2}[1 + cos(\beta_1 L_1 - \beta_2 L_2)]$$

Where I_o is the output intensity, I_i is the input intensity, $\beta_1 = \frac{2\pi n_1}{\lambda}$ and $\beta_2 = \frac{2\pi n_2}{\lambda}$.

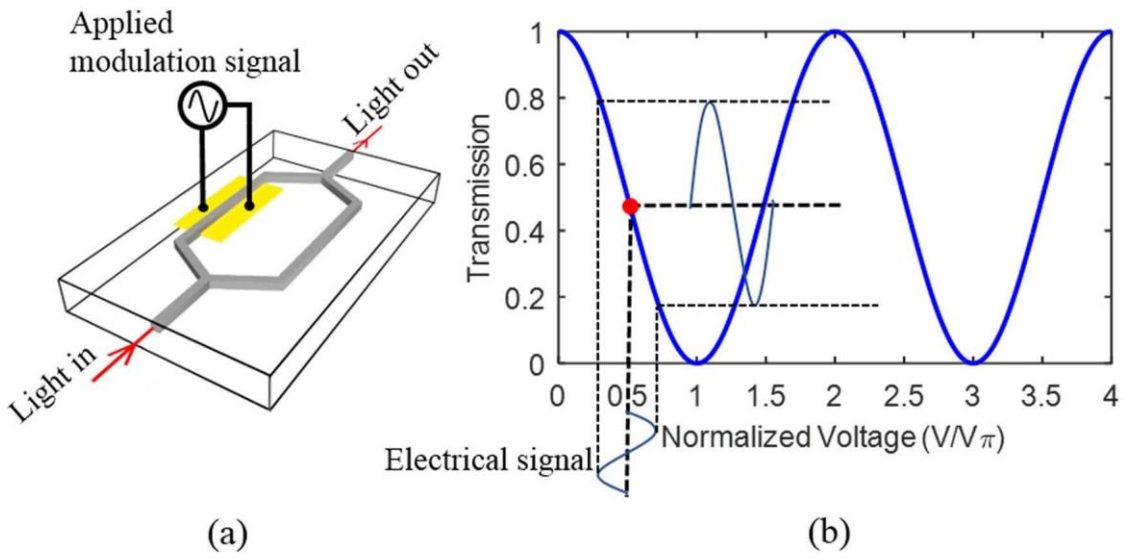


Figure 2.6: MZI modulator (a) MZI illustration, (b) MZI optical transmission as a function of normalized voltage.

2.5.3 Performance Metrics

In order to compare different modulators and how they operate, several metrics are used to characterize a modulator: optical bandwidth, modulation depth, footprint, insertion loss, and power consumption.

The definition of the optical bandwidth is as follows the Full-Width Half Maximum (FWHM) at the central wavelength of operation. Figure 2.7 shows a diagram of FWHM and how it is defined. Usually, the optical bandwidth is higher than the RF bandwidth (RF is the signal that needs to be modulated). The modulation speed is commonly characterized by its ability to carry data at specific rates. Interconnect application where high data transfer speed is required, a high modulation speed modulator is needed, while other applications such as sensing need relatively moderate speed modulators.

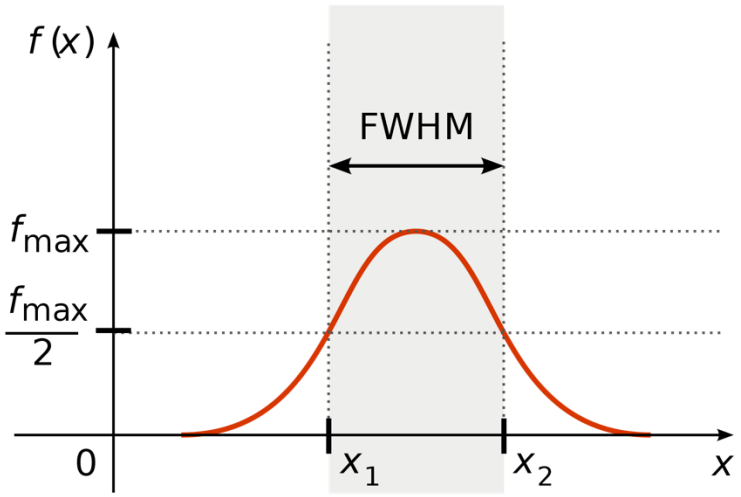


Figure 2.7: FWHM illustration.

Insertion loss is the optical power lost when the modulators are connected to the photonic circuit. It is the total loss of reflection, absorption, scattering, and mode coupling losses. The insertion loss is an essential aspect to be looked at, as the more loss you have, the lower optical power will reach the receiver. Silicon photonics has been in a trend to move toward smaller micrometer waveguides. Thus, the performance of the device is improved, but at the same time, the insertion loss increases. Another factor affecting the insertion loss is the sidewall angle and roughness. As waveguides are getting smaller in size, the sidewall roughness has an even more significant effect of increasing the scattering loss of the waveguide. Therefore, the performance of the modulators should be evaluated against any increase in the losses.

Another critical aspect of the device is the footprint. RR based modulators tend to be smaller in footprint compared to MZI based modulators [49], [55]–[58]. The bigger the footprint required by the modulator to higher, it suffers from cost, insertion loss, and power consumption. Due to the footprint factor, RR based modulators are superior to MZI based. However, other factors should be taken into consideration, such as the optical bandwidth and temperature stability.

Power consumption is an important metric to take into consideration when comparing modulators. The energy used to produce a bit of data is known as “energy per bit” or “power per bit.” To make optical interconnection feasible in comparison with electrical interconnection, the power per bit used to produce a bit of data in optical interconnection should not exceed the power used by electrical interconnection to produce a bit of data. In other words,

the energy per bit for future systems should be less than 1 pJ/bit [59]. Some authors suggested that in order to maintain this constrain, the total on-chip power needs to target around 100 fJ/bit to be more efficient than the electrical interconnects at the 45 nm process node [60]. Some typical electrical interconnects consumption of servers at present is of the order of 10-30 pJ/bit. In the present time, some authors have presented an electro-absorption plasmonic Graphene-based modulator integrated into silicon photonics consuming as low as 110 aJ/bit [61].

$V_\pi L$ is the product of voltage required to be applied on a waveguide to have a π phase shift in the optical response, typically it has a unit of V.cm. The lower $V_\pi L$ the better as the power consumption is lower. The following equation obtains this value [52]:

$$\Delta\phi(V)[\pi.cm^{-1}] = \frac{0.02\,\Delta n_{eff}(V)}{\lambda}$$

Where $\Delta n_{eff} = n_{eff}(V) - n_{eff}(0)$, λ is the wavelength in micrometer. Using the above equation, a phase change as a function of applied voltage (reverse voltage) is obtained and $V_\pi L$ is the value of the voltage applied at $\Delta\phi$ = 1. Figure 2.8 shows the change of effective index and α as a function of voltage, in this example the following parameters where used: waveguide width w = 500 nm, waveguide thickness t_{WG} = 220 nm, and slab thickness t_{slab} = 90 nm. The change in the phase is plotted in Figure 2.9. The voltage needed to have one π phase shift is 1.6 V and it is called V_π. Which mean that 1.6 V is needed to make a phase shift of π for a waveguide of 1 cm long.

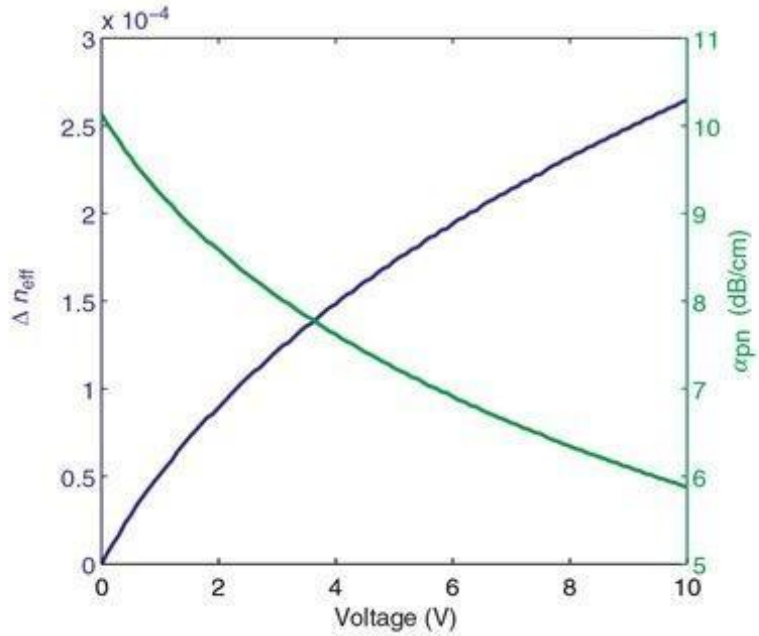


Figure 2.8: Change of effective and loss as a function of voltage (reverse biased depletion mode)[52].

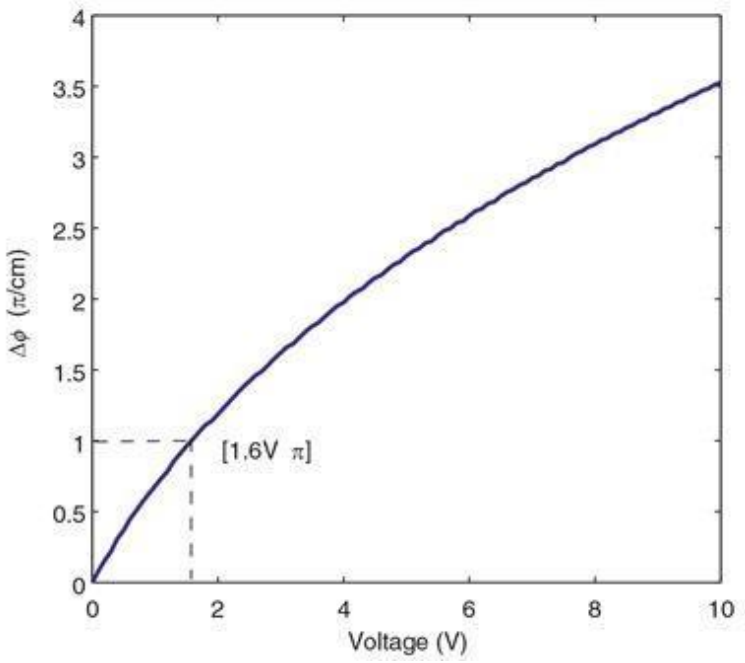


Figure 2.9: Phase change of a doped waveguide as a function of voltage (reversed biased depletion mode)[52].

The extinction ratio (ER) is defined as the ratio of the maximum intensity I_{max} (which is the maximum intensity of modulated transmitted light) to the minimum intensity I_{min} (which

is the minimum intensity of modulated transmitted light). Usually it is calculated in decibels (dB) as:

$$extinction\ ratio = 10\ log\ log\ \frac{I_{max}}{I_{min}}$$

The extinction ratio is also known as modulation depth. For long-distance transmission, a high extinction ratio is preferred. A ratio larger than 12 dB is preferred, especially when PAM4 modulation is used, this will allow having a 3-dB difference between the four levels of modulation in PAM4. Thus, a good eye diagram can be optioned using this ratio.

All these factors play an essential role in modulator design. A state-of-the-art comparison of different types of modulators is conducted to observe the advance in modulator research presented by different groups. Table 2.6 compares between different modulators in terms of $V_\pi L$, Data rate, Modulation, extinction ratio, and other factors.

Table 2.6 State of the art comparison of proposed modulator designs with different parameters.

Reference Paper	Data rate (Gbit/s)	Modulation	Extinction ratio (dB)	Central wavelength (nm)	$V_{\pi}L$ (V.cm)	Energy per bit	Platform	Year
[62]	40	NRZ	6.5	1550	11	4.5 pJ/bit	Academic Research	2011
[63]	53.5	PAM-4	3	1310	5.7 – 6.5	-	-	2014
[64]	25	-	6	1310	-	-	PIC25G + 65 nm CMOS	2015
[65]	10	-	8	1540	-	-	BiCMOS	2013
[66]	56	-	4	1566	-	45 fJ/bit	300mm	2015
[67]	25.6	PAM-4	10	1310	-	-	-	2015
[68]	200	16-QAM	-	-	-	-	-	2016
[69]	20/40	NZR/PAM-4	3	1280	-	0.15/0.042 pJ/bit	45 nm CMOS	2017

[70]	25/56	NZR/PAM-4	6.3	1300	-	4.8 pJ/bit (PAM-4)	90 nm CMOS SOI	2016
[71]	40	PAM-4	7	1550	-	3.04 pJ/bit	130 nm SOI	2016
[72]	10	NZR	6	-	-	1.27 pJ/bit	65 nm CMOS	2016
[73]	100	PAM-4	-	1550	Short Segment 2.5 Long Segment 2.7	-	-	2015
[74]	100	PAM-4	20	1550	5.5	-	-	2017
[75]	50	NZR	5.6	-	2.4	-	130nm CMOS	2018

2.6 Design Challenges

Designing a modulator is a challenging task as it involves a trade-off between all the design parameters: optical bandwidth, optical loss, power consumption, modulation efficiency, thermal stability, and footprint. For instance, one of the most fundamental design parameters is doping concentrations. High doping concentration means a high number of free charge carriers, thus, a high modulation efficiency. Nevertheless, at the same time, the optical loss is increased by the high doping concentration. In order to solve this dilemma, the doping can be manipulated to reduce the loss as follows: active part (around the depletion region) is doped with high concentration, while the other part of the active region is lightly doped or remain intrinsic [76]–[78]. Figure 2.10 shows a representation view of the proposed doping with different concentrations. By applying this doping profile, the optical loss can be reduced dramatically, while maintaining a high modulation efficiency. The linearity of the devices can be achieved by using this doping profile [78] as the depletion region can be confined inside the high concentration region. Also, as mentioned in the previous section, it is preferable to have more p-type as the contribution of holes on changing the refractive index is high, while it has a low contribution in increasing the absorption. This means shifting the junction position to the n-type doped side is better as some researchers have demonstrated a performance improvement by using this offset [79], [80].



Figure 2.10: proposed doping profile with higher doping near the depletion region.

Designing an ultra-high-speed athermal modulator is challenging. Also, the design needs to minimize the following modulator parameters: footprint, the effect of temperature change on modulation, power efficiency, and increase extinction ratio. The optical and electrical study is needed in order to fulfill the requirements of the design.

Finally, one of the main challenges in designing modulators (MZM and MZI-RR modulators) is the lack of a unified simulation platform tools for electrical and optical to be simulated together. Optimizing all of the parameters of the design in an integrated electrical, optical simulation tool is required to achieve the highest performance form the device. Lumerical simulation software integration between the different parts of the simulation, but each simulation is independent of the other. While this is a good starting point for simulation software, the demand for the high-speed optical link will encourage simulation providers to develop and improve their tools.

Chapter 3: Athermal Modulator design

3.1 Key parameters for athermal design

Semiconductor materials have a different refractive index (which is explained in the previous section); similarly, a waveguide has a unique, effective index which describes the ratio of the speed of light in the waveguide relative to vacuum for a given polarization, wavelength, mode of propagating light and waveguide dimensions. The effective index of a waveguide is described using the following equation:

$$n_{eff} = \frac{\beta}{k_0}$$

Where β is the propagation constant and k_0 is the wavenumber ($k_0 = \frac{2\pi}{\lambda_0}$). In order to fully understand the effective index, a couple of parameters affecting the index required to be explained.

Electromagnetic waves (light in this thesis) are waves that consist of electrical and magnetic fields traveling along the axis of propagation and orthogonal to each other. Figure 3.1 shows an example of the electromagnetic wave travels along the z-axis. Thus, light is propagating through a waveguide in transverse electric known as TE mode or transverse magnetic known as TM mode. TE mode is a mode where the electrical filed E is always perpendicular to the direction of propagation, while TM mode is a mode where magnetic field B is always perpendicular to the direction of propagation. In a waveguide, light can propagate in different modes such as TE_0, TM_0, TE_1, TM_1.

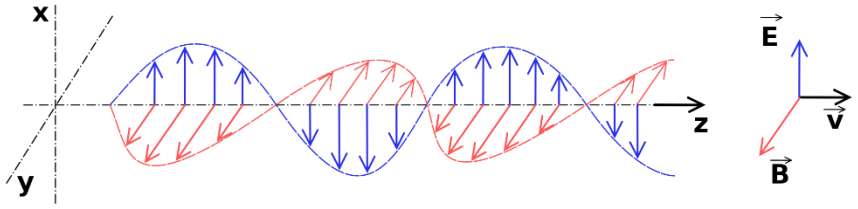
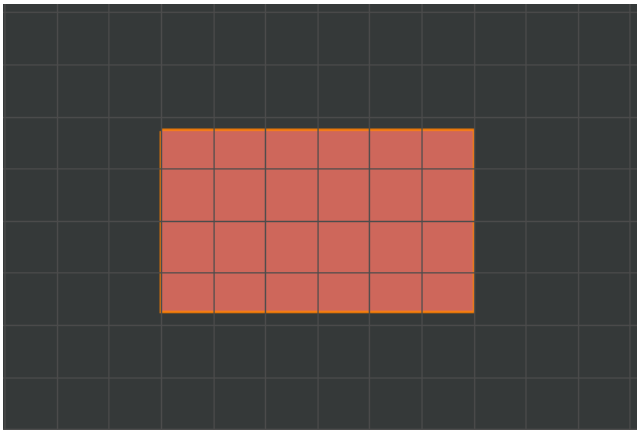


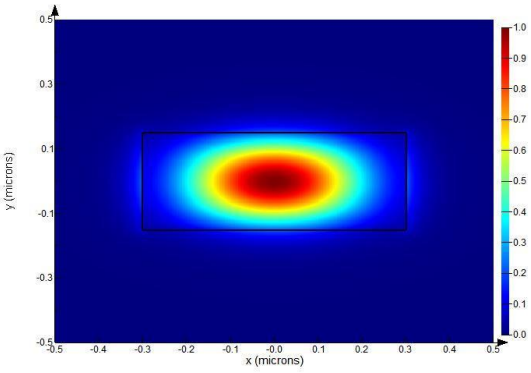
Figure 3.1: illustration of electromagnetic wave.

The number of modes that a waveguide (dielectric structure) supports depends on the dimensions of it.

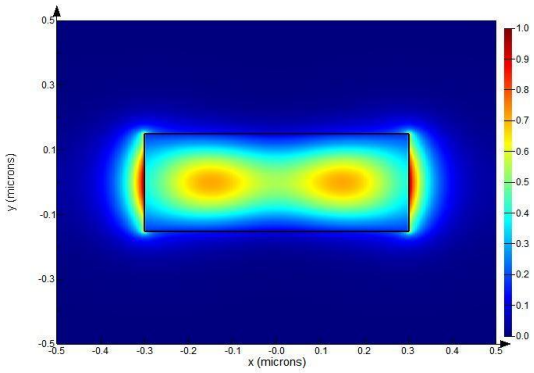
As waveguides have finite dimensions, the rectangle dielectric waveguide theory is essential to be taken into consideration. The theory suggests that there is a finite number of modes the waveguide can support, and it is related to the dimensions of the waveguide. Figure 3.2a shows an example of a silicon waveguide with silicon dioxide as its cladding with the following dimensions: width = 600 nm and thickness = 350 nm. Simulating the waveguide using Lumerical mode at a wavelength of 1310 nm, the waveguide supports multimode TE and TM. Figure 3.2b-e shows the different modes supported at that wavelength.



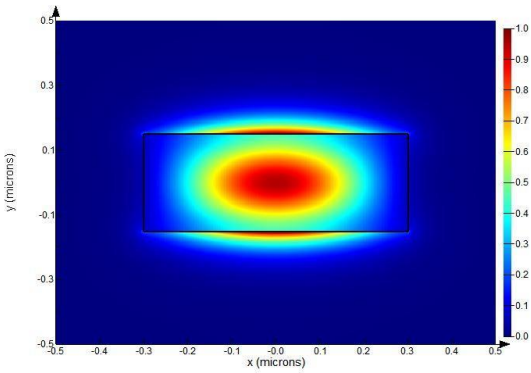
(a)



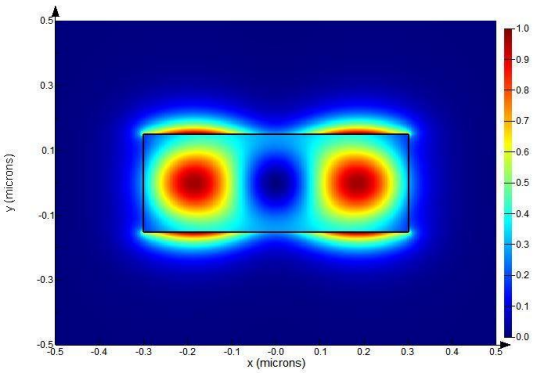
(b)



(c)



(d)



(e)

Figure 3.2: Silicon waveguide with its supported modes (a) waveguide schematic (b) TE_0 mode, (c) TE_1 mode (d) TM_0 mode (e) TM_1 mode.

Group index is also known as group refractive index n_g is the ratio the speed of light in a vacuum to the group speed in the material:

$$n_g = \frac{c}{v_g} = n_e - \lambda \frac{dn_e}{d\lambda}$$

Where c is the speed of light in vacuum, v_g is the group speed or know as group velocity, n_e is the effective index, and λ is the wavelength. The importance of group index lies in that the light is not a single wavelength (single pure color). Thus, the group index describes how such a light will propagate through a medium or a waveguide in this case. Generally, the group index is higher than the effective index, which means the whole pulse (which is not a single wavelength) will travel slower than a single wavelength in the same waveguide. This reason leads to a pulse spread out during the travel process.

The thermal-optical coefficient (TOC) is an essential property of silicon, as it is affecting the refractive index of silicon. TOC has been measured to be $\frac{\partial n}{\partial T} = (1.86 \pm 0.08) * 10^{-4}$ (K^{-1}) at a wavelength of 1.5 μm [5]. Due to the high TOC of silicon, an athermal design of a modulator where it can modulate on a wide range of temperatures is critical to be achieved, and this is the core study of this thesis.

3.2 Design concept

Designing an athermal modulator is a challenging task because many trade-offs need to be taken into consideration. Research groups have proposed different paths to follow when designing an athermal modulator such as Athermal MZI by carefully choosing the dimensions of the modulators to make a passively compensate for the temperature change [15], Another

way is to have a ring resonator to stabilize MZI and create athermal modulator [80] thermally. In this thesis



Figure 3.3: MZI schematic.

both the concepts will be analyzed, studied, and simulated to design an athermal modulator for GF-45SPCLO technology and GF-90WG technology.

MZM is sensitive to temperature change due to the high TOC of silicon. The optical path difference (OP) in MZM is described by: $n_2L_2 - n_1L_1$, where n_1, n_2, L_1, L_2 are the effective index of the waveguide of the first arm, the effective index of the waveguide of the second arm, first arm length, and second arm length, respectively. Figure 3.3 demonstrates a schematic diagram of the MZI. Theoretically, if there is no change on optical path difference with temperature ($\frac{d\,\Delta OP}{dT} = 0$, where ΔOP is the change in optical path difference, and T is the temperature in K) the device should be athermal. This is because the optical path is the effective path that the wave sees when propagating through a waveguide, and if there is no change in the optical path with temperature, the wave will not face any change. Thus, the output behavior is thermally stable.

3.2.1 Athermal MZI design concept

Following the concept proposed by J.Viegas [15] to design an athermal MZI modulator, a specific condition needs to be achieved:

- The spectral shift with temperature should be zero.
- The free spectrum range (FSR) change with temperature should be zero.
- The dependence of temperature sensitivity on wavelength should be zero.

Translating the conditions mentioned above into mathematical equation alongside the derivation of the equations was fully described in [15]. The following equation is needed to be solved for each GF technology (GF-45SPCLO and GF-90WG) in order to design an athermal MZI [15].

$$\frac{\partial n_1}{\partial T}L_1 - \frac{\partial n_2}{\partial T}L_2 - \frac{\partial n_3}{\partial T}L_3 = 0$$

$$\frac{\partial^2 n_1}{\partial T \partial \lambda}L_1 - \frac{\partial^2 n_2}{\partial T \partial \lambda}L_2 - \frac{\partial^2 n_3}{\partial T \partial \lambda}L_3 = 0$$

$$FSR = \frac{\lambda^2}{n_{g,1}L_1 - n_{g,2}L_2 - n_{g,3}L_3}$$

Where n_1, $n_{g,1}$, L_1 are effective index, group index, and arm length, respectively. To solve those equations full optical simulation analysis is needed for the technology stack by setting some parameters and solve for the arm lengths, which will make the modulator athermal.

The first step of designing the athermal modulator is to set the waveguide profile, which means choosing the material and geometry. As this design is tolerated for GF and their technologies, the layer stack will be taken from GF-45SPCLO and GF-90WG technologies. The waveguide core is silicon, and the fabrication process provided by GF fixes the thickness

of the waveguide in the range of [155-165] nm, so the only variable that can be manipulated with is the width w of the waveguide. Starting with a simple approach for an MZI modulator is to have the first arm length L_1 and waveguide width w_1 and the second arm length L_2 and waveguide width w_2. The input light is assumed to be traveling in TE_0 mode; to get an athermal modulator it is required to solve for the above equations as described above. If $w_1 = w_2$ both the waveguides will have the same effective index and $\frac{\partial n_1}{\partial T} = \frac{\partial n_2}{\partial T}$, thus will lead to $L_1 = L_2$. Moreover, since both arms have the same lengths, the FSR requirement cannot be met. When $w_1 \neq w_2$ and following the same procedures explained with the previous case, all the conditions can be met, but the FSR will be fixed, and it cannot be changed. Finally, to have control over FSR a third waveguide must be introduced with a width w_3 and length L_3. Then by solving the equation for known widths w_1, w_2, w_3 a finite solution can be found for a set of L_1, L_2, L_3 and the FSR could be controlled by changing the $L_1, L_2\ and\ L_3$.

Mathematically, for any combination of waveguide widths w_1, w_2, w_3 there is a solution to $L_1, L_2\ and\ L_3$, and a different set of width will give different sets of arm lengths, which will result in a different footprint, which is preferred to be minimized. To minimize the footprint of the modulator different set of waveguide widths alongside with their different propagation modes and waveguide properties ($\frac{\partial n}{\partial T}$, $\frac{\partial^2 n}{\partial T \partial \lambda}$, n_g) is used to optimize the arm lengths to get the smallest athermal modulator possible.

A schematic of the proposed athermal MZI [15], which used SOI as a technology platform to develop the modulator is shown in Figure 3.4. In the proposed design, two modes where used TE_0 as the input waveguide and TE_1 in the other arm to minimize the footprint of the device. In order to ensure having a functional coupling from the fundamental mode to the

first-order mode in TE, the effective index n_{eff1} at w_1 and TE_0 must be equal to n_{eff2} at w_2 and TE_1 which is known as a phase matching condition. The final waveguide width was chosen to minimize the footprint of the device, and the proposed dimensions are shown in Table 3.1.

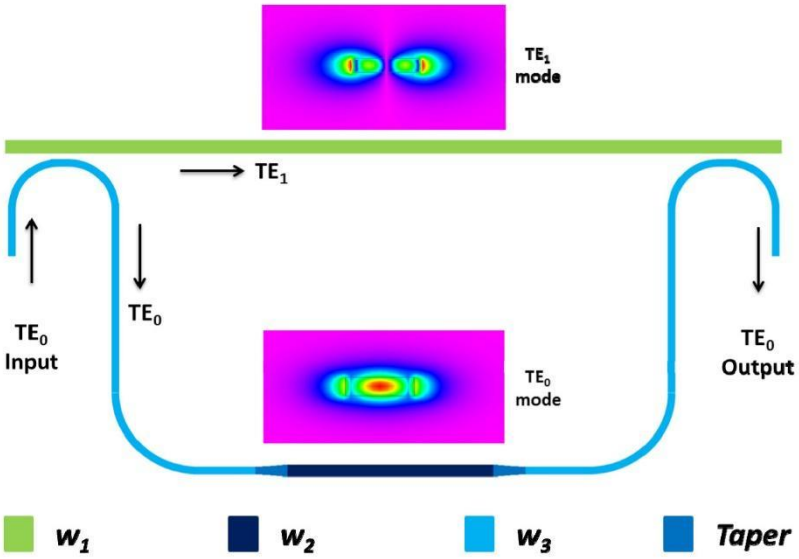


Figure 3.4: Proposed athermal MZI schematic with TE mode profile in each arm.

Table 3.1: Proposed waveguide length of athermal MZI by [15].

Arm	Mode	Waveguide width (nm)	Waveguide length (µm)
1	TE_1	600	294.6
2	TE_0	400	25.8
		270	374.6

3.2.2 Athermal MZI-RR design concept

RR can passively compensate for the change in temperature of MZI. This combination between MZI and RR will enable the creation of ultralow-power optical interconnects. Micro ring resonators offer a significant advantage that lies in their high extinction ratio, compact size, low power consumption , and low insertion loss [49], [81], but the RR is highly sensitive to temperature due to their narrow optical bandwidth. In order to compensate for this high sensitivity of temperature, the concept design structure consists of a ring resonator over-coupled to a balanced MZI [15], [82], [83]. The design concept schematic of the MZI-RR athermal modulator is shown in Figure 3.5a. Using a similar approach described in the athermal MZI design concept and as the design is created for GF-45SPCLO and GF-90WG technologies, the additional degree of freedom in the design is the waveguide widths [84], apart from the arm lengths. The parameters of the MZI (widths and arm lengths) are chosen in a way to compensate for the redshift of RR with temperature, Also to make MZI optically balanced ($Optical\ Path\ difference_{MZI} = 0$). Figure 3.5c shows how the behavior of MZI and RR should look like with the correct chosen parameters to make the device thermally stable. MZI should have strong negative temperature sensitivity $\left(\frac{\partial}{\partial T}(Optical\ Path\ difference_{MZI}) < 0\right)$. In the other hand the sensitivity of the RR should be highly positive $\left(\frac{\partial}{\partial T}(Optical\ Path\ difference_{RR}) > 0\right)$ with relatively large waveguide width to enable highly confined single mode operation.

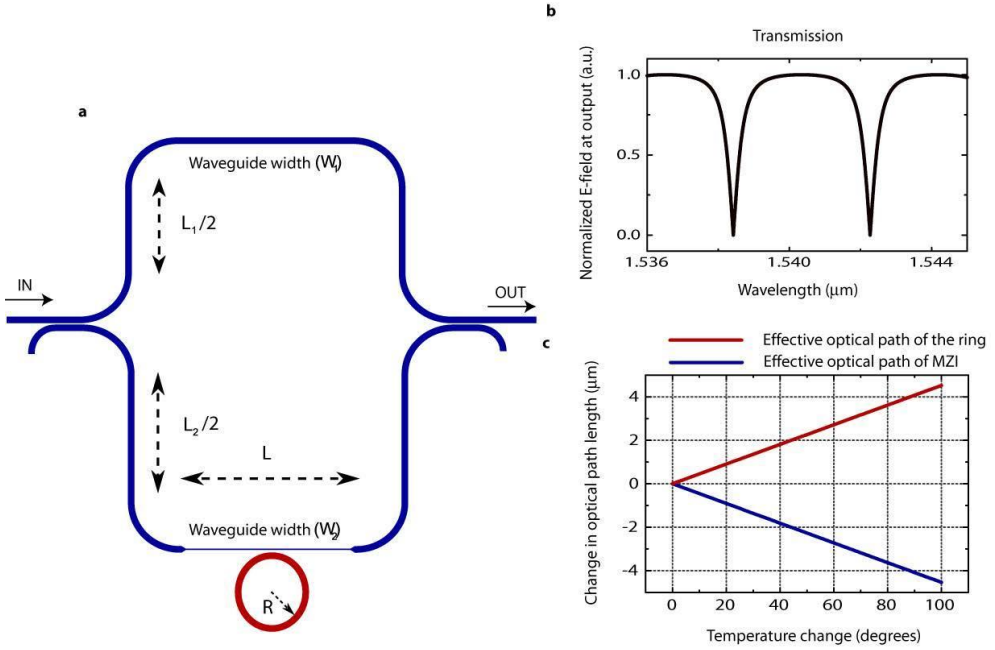


Figure 3.5: (a) Concept MZI-RR design schematic with various lengths and widths. MZI is shown in blue and RR in red. (b) The transmission spectrum of a 40 μm ring radius with balanced MZI (the optical path length is equal in both arms). (c) Change in optical path length as a function temperature for RR and MZI *[85].*

The advantage of this design concept is that the resonance oscillates about the central wavelength alongside with temperature instead of shifting away with typical RR, this is because of the interplay of the MZI (blueshift with temperature) and the RR (redshift with the temperature). The phase change in MZI is linear, while the phase change in RR is non-linear, and thus the oscillations occur. The phase difference is described by the following equation [83], [85].

$$\frac{\partial}{\partial T}\phi_{REMZI}(T) = \frac{\partial}{\partial T}\left\{phase\left(\frac{t-\alpha e^{j\beta L_{Ring}(T)}}{1-\alpha t e^{j\beta L_{Ring}(T)}}\right)\right\} + \frac{\partial}{\partial T}(\beta L_{MZI}(T)).$$

Where L_{MZI}, L_{Ring} are the optical path length of the MZI and RR, respectively, $(1-\alpha)$ is the roundtrip loss in the ring, t is the coupling coefficient between the MZI waveguide and RR waveguide, and β is the propagation constant.

Compensating the RR temperature phase shift is only occurring with a particular choice of compensating MZI to result in perfect oscillations. If the compensation is too low or too high, the response will drift away with temperature. To mathematically solve for the MZI it is divided into two sections: In the first section, each arm has different waveguide widths, but the same arm lengths L_{MZI}. In this case, the section is characterized by the difference in the effective indices of the arms Δn_{eff}. In the second section, the two arms have the same waveguide widths n_{eff}, but different arm lengths, thus this section is characterized by the difference in the arm lengths ΔL_{MZI}. Figure 3.6 shows a schematic of the different sections of the device. The RR is over-coupled to the arm with a thinner waveguide width of the MZI. The optical path shift with the temperature of the MZI and RR are equal when the following condition is satisfied.

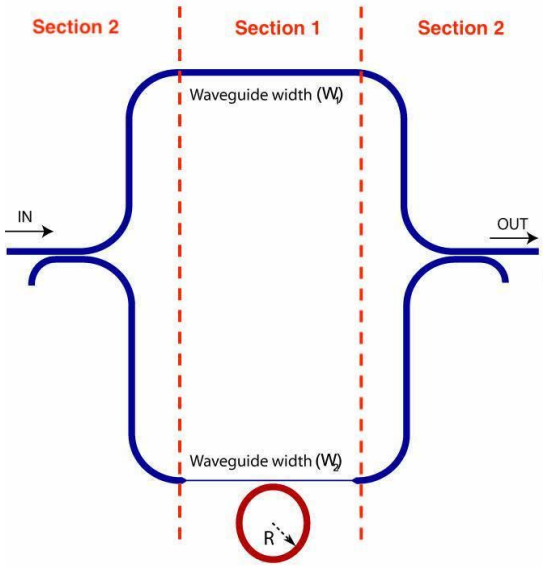


Figure 3.6: MZI-RR athermal modulator concept design sections [85].

$$MZI\ output\ (T) = \left[n_{eff} + \frac{\partial n_{eff}}{\partial T}T\right]\Delta L_{MZI} + \left[\Delta n_{eff} + \frac{\partial \Delta n_{eff}}{\partial T}T\right]L_{MZI}$$

$$= -\left[n_{eff} + \frac{\partial n_{eff}}{\partial T}\right]2\pi R$$

[85] Where R is the ring radius that the MZI will compensate for the temperature. The athermal MZI is designed in a way to compensate the RR phase shift, by equating coefficients of T and solving the equation to get ΔL_{MZI} and L_{MZI}.

3.3 Athermal MZI design in GF-45SPCLO technology

GF-45SPCLO technology stack used for designing an athermal modulator. In order to correctly design the modulator, a full simulation analysis needed to be done to understand the limitation in this technology and how it can be used in the design process. GF gave the stack layer, and the thickness of each layer cannot be changed as this is part of the fabrication process at GF. In designing the modulator, a slab waveguide is required, so the slab can be used later to control the voltage across the modulator. Figure 3.7 shows a cross-section of the slab waveguide used to design the modulator; the only degree of freedom available in the design process is the waveguide width. The core of the waveguide is Silicon, and the TOC of silicon is about $(1.86 \pm 0.08) * 10^{-4}\ (K^{-1})$ [5]. The remaining layers data (refractive index and absorption coefficient) were provided by GF as part of the technology stack.

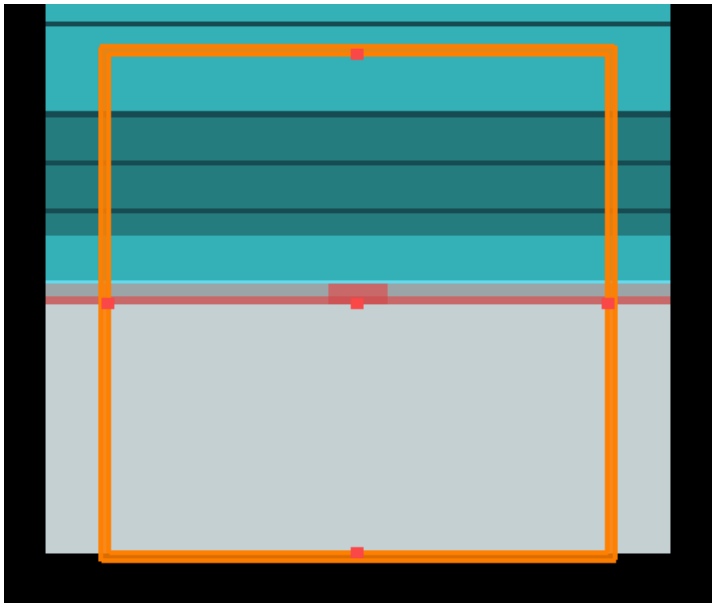


Figure 3.7: Slab waveguide cross-section used in GF-45SPCLO technology for modulator design.

In order to be able to design the modulator properly, a couple of simulations was required. The simulation of the design was conducted in two stages. Stage one which takes care of all the passive simulation and results related to it, such as the effective index of the waveguide, losses, group index, $\frac{\partial n_{eff}}{\partial T}$ and $\frac{\partial^2 n}{\partial T \partial \lambda}$. While the second phase took care of active simulation part such as doping profile, voltage characteristics, and $v_\pi L$. The simulation tools used during the designing process are Lumerical Mode (for passive simulation, optical response), Lumerical device (for active simulation, doping, and voltage) and Lumerical Interconnect (Full system simulation). Also, MATLAB was used to analyze and interpret the data gathered from different simulation tools.

3.3.1 Simulation setup

As mentioned previously, GF-45SPCLO technology provided the information required for all the layers above and below the silicon layer (the waveguide layer), while in the

simulation the following material was used for silicon and silicon dioxide: (Si) Silicon – Palik and SiO_2 (Glass) – Palik respectively. That material is found in Lumerical software material database. The central wavelength used in the simulation is 1310 nm (O-band). To get the required parameters for the design couple of parametric sweeps were simulated and the gathered as follows:

- Effective index as a function of waveguide width (width varies from 200 nm – 800 nm).
- Effective index as a function of wavelength (Frequency varies from 228 THz – 238 THz) (a wavelength of 1260 nm – 1310 nm).
- Effective index as a function of Temperature (temperature varies from 300 K – 380 k).

By sweeping the parameters mentioned, and gathering the effective index data, a full study of silicon behavior in this technology was obtained. These sweeps are important to be able to understand the behavior of the waveguide as the temperature change and to use this behavior in thermally stabilizing the device. Another aspect of the athermal MZI needs to be taken into considerations such as: Bend radius simulation (the smallest bend radius where the loss is minimum), input coupler (the input coupler where the light will be coupled from the fundamental mode to the first mode), output coupler (where the interference will occur after the light (electromagnetic wave) rejoin together) and waveguide tapper (two types of tappers will be studied. First is the tapper that changes from fully etched waveguide to slab waveguide, the second tapper is the tapper that change form one waveguide width to another). All of these simulations and studies will be conducted to design the athermal MZI.

3.3.2 Waveguide widths and Arm lengths

Figure 3.8 shows the modal profile of the waveguide at a wavelength of 1310 nm. The waveguide is operating in a single mode when the waveguide width < 350nm, as shown in Figure 3.8a, and the first-order modes start to appear when width > 350nm, as shown in Figure 3.8. This information is important as per the design concept explained earlier. It is required to have a waveguide supports TE_0, and another supports TE_1. Using the modal profile information gathered from the Mode simulation, it is possible to divide the simulation into two parts, where the first part the parametric sweeps mentioned earlier is conducted for the fundamental mode (TE_0) and the second part the same sweeps conducted for the second mode (TE_1). Both the data generated by the parametric sweep where gathered, analyzed, and studied together to be used in the design process.

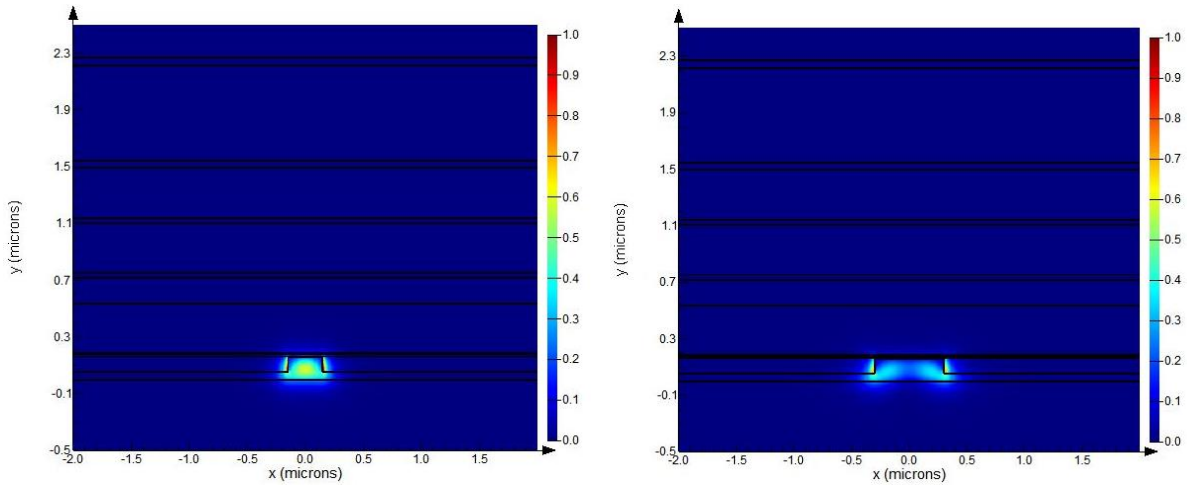


Figure 3.8: GF-layer stack modal profile at a wavelength of 1310 nm (a) Fundamental TE mode (b) First order TE mode.

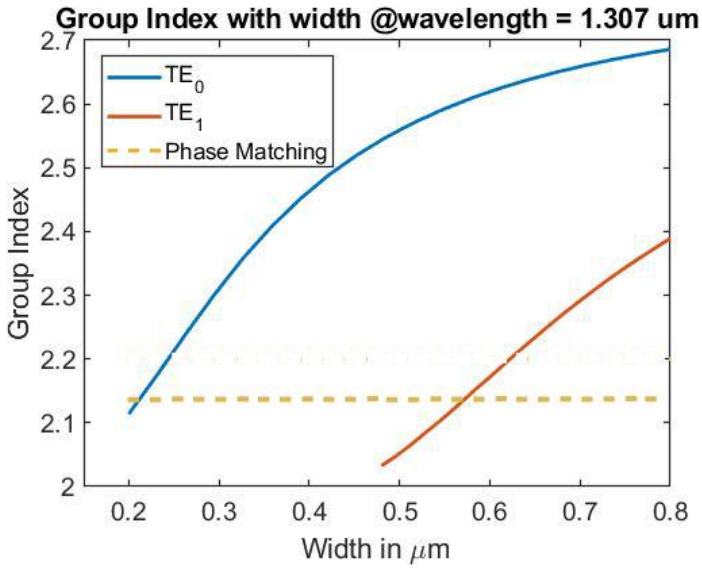


Figure 3.9: Group Index of Silicon waveguide in GF-45SPCLO technology as a function of waveguide width.

Figure 3.9 plot the group index as a function of waveguide width. The blue line represents the fundamental mode (TE_0), while the orange line represents the first-order mode (TE_1). This plot is used to find the phase-matching widths by fixing the group index and scan across the x-axis (width of the waveguide) to get two different widths that can lead to the same group index. Phase matching condition is essential as it is used in the input coupler of the athermal MZI design, as described in the design concept, the input light (electromagnetic wave) is launched into the input as a TE_0 mode and using the phase matching condition a coupler that couple TE_1 from TE_0 is achieved. Thus, the first two waveguide width is set by this condition, and they are w_1 = 558 nm, w_2 = 203 nm, and this for a group index of 2.12.

Temperature affects the refractive index of the material, hence affects the effective index of the waveguide. Figure 3.10 shows the effective index is changed by the temperature as a function of waveguide width. Figure 3.11 shows the effective index changed by temperature and wavelength as a function of waveguide width. Those two figures are used to

get the values of $\frac{\partial n}{\partial T}$ and $\frac{\partial^2 n}{\partial T \partial \lambda}$ at each waveguide width. Using the different constant found at each waveguide width and setting the required FSR, then substituting in the equation the arm lengths required to make the MZI athermal is found. This process is repeated multiple times with different values for w_2 to find the smallest athermal MZI footprint, as different sets of waveguide width will lead to different sets of arm lengths. This calculation process was done using a MATLAB code where it will look for the required values at each waveguide width, store the value and then use them in solving for the arm lengths after multiple repeating the values of waveguide width and arm lengths produced the smallest footprint athermal MZI is shown in Table 3.2.

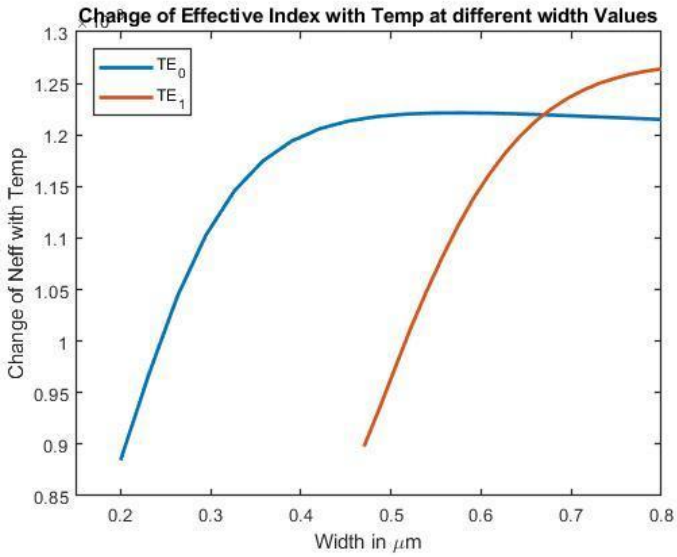


Figure 3.10: Change of effective index as a function of waveguide width for GF-45SPCLO technology.

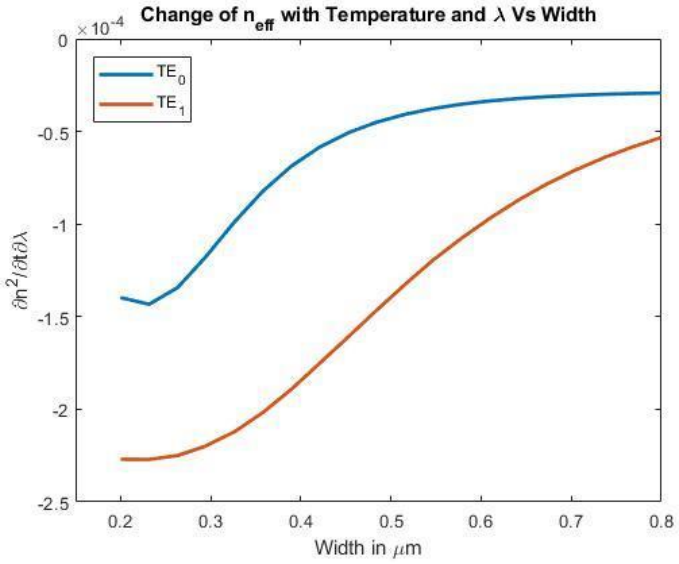


Figure 3.11: Change of effective index with temperature and wavelength for a waveguide width for GF-45SPCLO technology.

Table 3.2: Waveguide widths and arm length for GF-45SPCLO athermal MZI.

MZI Arm	Operational mode	Waveguide width (nm)	Arm Length (μm)
Upper Arm	TE_1	$w_1 = 558$	1838.9
Lower Arm	TE_0	$w_2 = 400$	1113.5
		$w_3 = 203$	1020.6

3.3.3 Bend radius

Typically, waveguides are connected using bends to interconnect the dielectric waveguide pathways [86]. Bends are the basic building block in many optical devices such as MZI, coupler, and RR, but the bent waveguide can increase the loss and reduce the performance

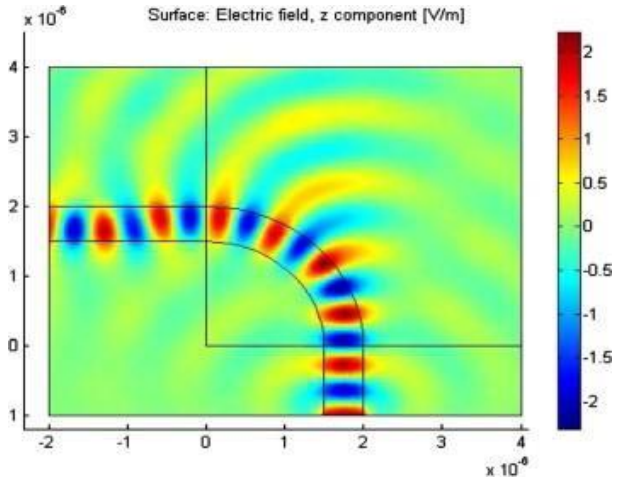
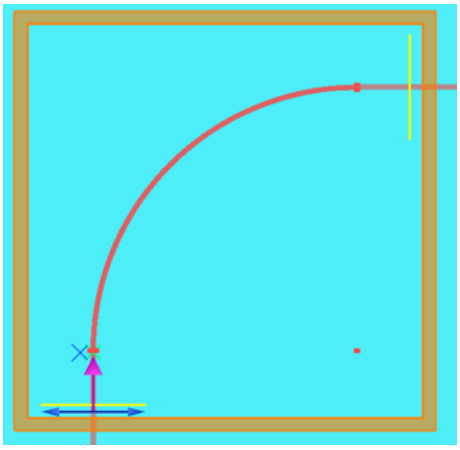


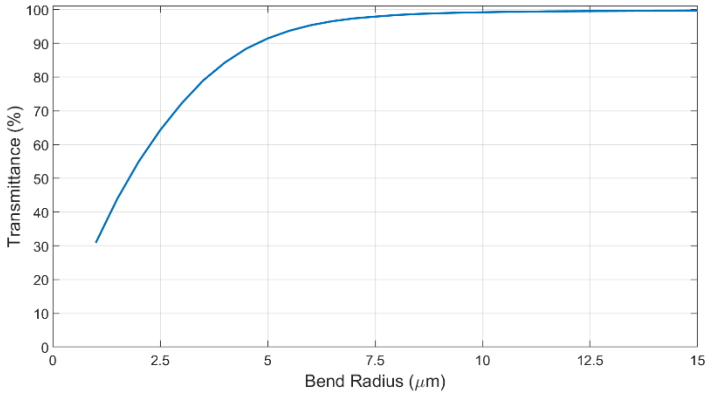
Figure 3.12: waveguide bend with radiation loss due to the bend.

of the device. Mainly the loss increases due to the radiation loss, where the light (electromagnetic wave) will radiate out of the waveguide, and this loss is directly proportional to the bend radius. Figure 3.12 shows a waveguide bend, and the radiation loss occurs due to it. Also, the loss is different with each mode of propagating in the waveguide. The first-order mode has a greater loss than the fundamental mode; due to this fact, the bend study is conducted by using the first-order mode. Several geometries proposed to create the S-shaped bending waveguide, such as back-to-back curvature and sinusoidal S-bend [87].

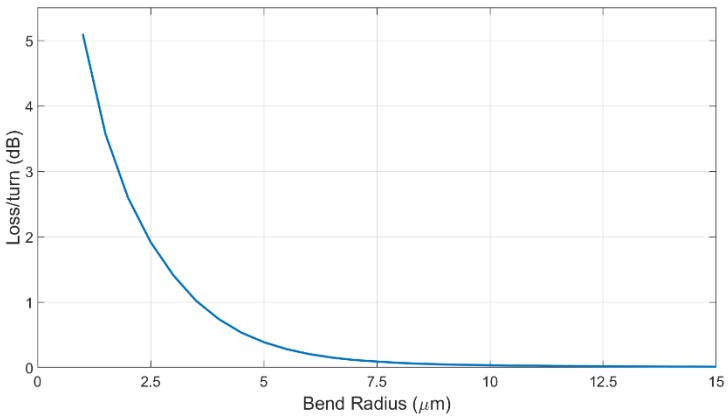
Figure 3.13a shows the simulation setup using Lumerical Mode solver used in the study, where R is the bend radius used as a changing variable to observe the transmission and loss (loss per turn for a 90- degree bend and propagation loss) for different radiuses. As it is shown in Figure 3.13b, with the increase of radius, the transmission increases, and it reaches 99.4% (percentage relative to the input light) with a radius of 15 µm. At the same radius, the loss per turn is almost zero Figure 3.13c, and the propagation loss is about 0.05 dB/cm Figure 3.13d. Using the data gathered from the bend simulation, a radius of 15 µm is used for all the waveguide bends in the design, including the input coupler and output coupler bends.



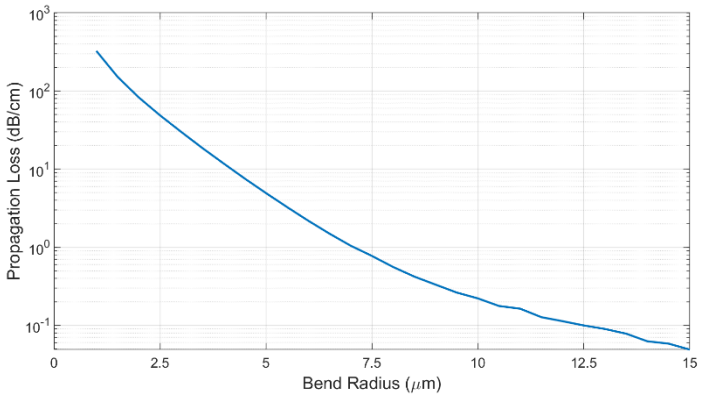
(a)



(b)



(c)



(d)

Figure 3.13: waveguide bend study (a) simulation setup (b) Transmittance as a function of bend radius (c) Loss per turn as a function of radius (d) Propagation loss as a function of radius.

3.3.4 Couplers designs

In photonic systems splitting and combining lights is done by using couplers, which is the most commonly used building block to do it. The coupler consists of two parallel waveguides that brought closer to each other to allow the light to be coupled from one to another. The coupling coefficient is controlled by the waveguide length and the gap separating

the waveguides. Couplers can be implemented using any type of waveguides (strip or rib). Figure 3.14 shows a schematic of a coupler. A coupler for two waveguides can be characterized by the coupling coefficient C and the cross-over length L_x. The cross-over length is the length required for light to be completely coupled from one waveguide to the other. In other words, the intensity of light in the second waveguide is equal to 100% when the cross-over length is reached. For a high-index contrast waveguide, the supermode approach known as the eigenmode expansion method is more accurate than the traditional Coupled Mode Theory [52]. The coupling coefficient C for two supermodes can be described using the following equation [52]:

$$C = \frac{\pi \Delta n}{\lambda}$$

Where $\Delta n = n_1 - n_2$ which is the difference between the effective indices, λ is the wavelength as the coupling coefficient is different for each wavelength. The cross-over length L_x is described by the following equation [52]:

$$L_x = \frac{\lambda}{2\Delta n} = \frac{\pi}{2C}$$

As it is obvious from the above equation, the cross-over length is inversely proportional to the coupling coefficient, so as the coupling coefficient increases, the length decreases.

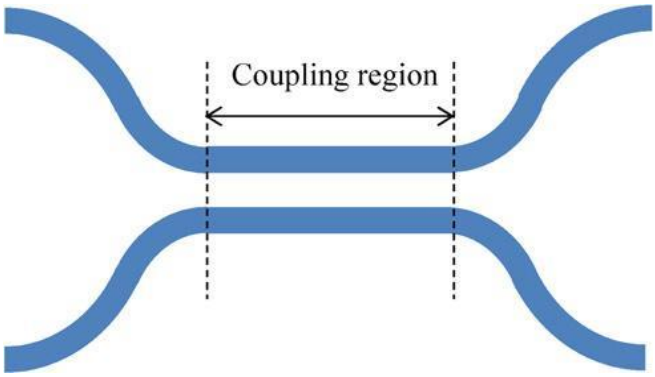


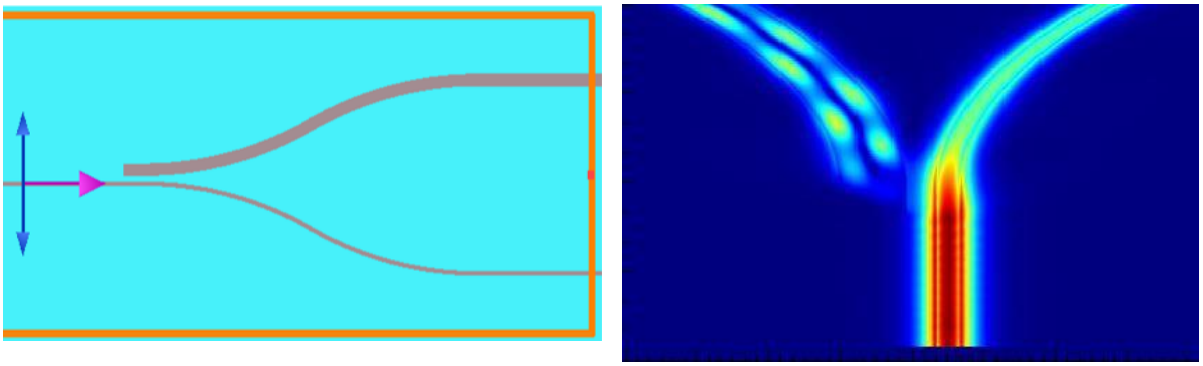
Figure 3.14: Directional coupler schematic.

3.3.4.1 Input Coupler design

The athermal modulator MZI concept requires the input light to be coupled to the first-order mode (TE_1) from the fundamental mode (TE_0). In order to fulfill this requirement, a simulation setup using Mode solver was required to find the optimum gap and length to split 50% of the light to the second waveguide, the simulation setup is shown in Figure 3.15a. Using a bend radius of 15 μm for bending the waveguide away after coupling the light Figure 3.15b shows how the light is coupled from the fundamental to the first-order mode with a splitting ratio of 50:50. Optimizing the power transferred from one waveguide to another is important to reduce the performance loss of the coupler design. This study was done by looking into different cases where the separation gap between the waveguide changes.

Table 3.3 shows the different parameters used alongside with the output power achieved is each case. Figure 3.16 shows the result obtained from the simulation for the different cases of the coupler, and it plots the power as a percentage as a function of coupler length where the blue line is the input power in TE_0 in the input waveguide, the red line is the coupled power in TE_1, and the green line is the coupled power in TE_0 in the second waveguide,

Table 3.3 summarizes the result obtained.



(a) *(b)*

Figure 3.15: Input coupler design (a) Simulation setup (b) Power profile of the simulated design.

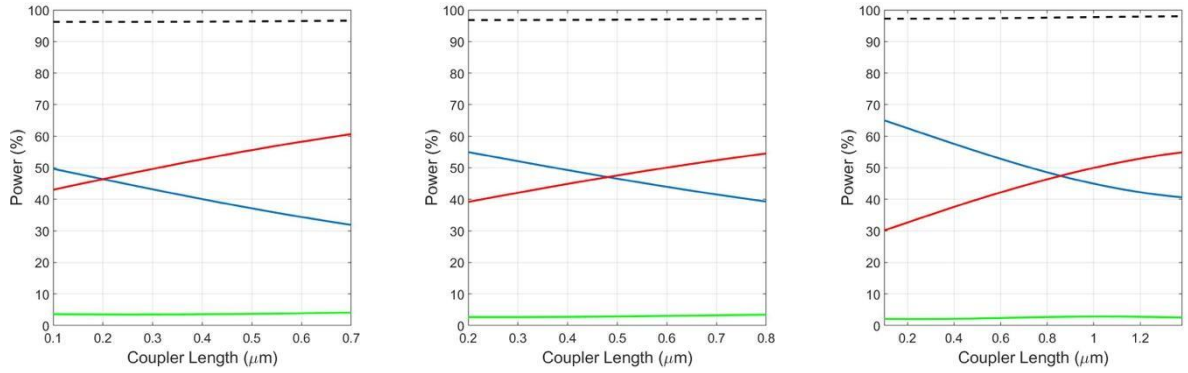


Figure 3.16: Input coupler power comparison with different gaps between the waveguides (a) gap of 200 nm, L3dB = 0.2 μm (b) gap of 225 nm, L3dB = 0.48 μm (c) gap of 250 nm, L3dB = 0.855 μm.

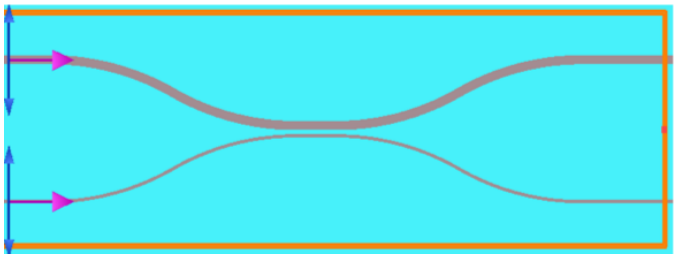
Table 3.3: Different gap separation couplers with power transferred in each waveguide and mode, power value is normalized to the input power.

Separation Gap (nm)	$L_{50:50}$ (μm)	First waveguide	Second waveguide		Lost power
		Power in TE_0	Power in TE_0	Power in TE_1	
200	0.2	46.37%	3.48%	46.37%	3.78%
225	0.48	47.06%	2.82%	47.06%	3.06%
250	0.855	47.4%	2.74%	47.4%	2.46%

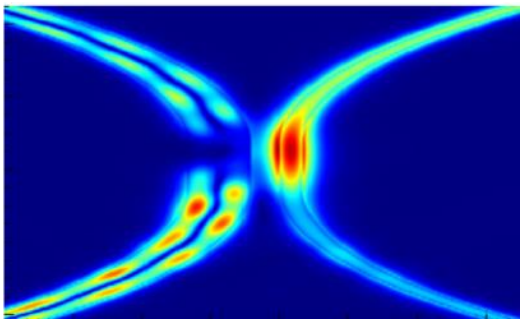
3.3.4.2 Output Coupler design

The same approach described in the previous section used for designing the output coupler. The main difference between the input coupler and the output coupler is that the input couples the light that comes from one waveguide with 100% power and it required to transfer 50% of the power to the second waveguide, while in the output coupler (where the interference occurs) the light is coming from both the waveguides with 50% in each arm. It is required that both lights interfere and separated again into two waveguides. The simulation is taking into consideration that one waveguide has light is the fundamental mode (TE_0), and the other waveguide light is in first-order mode (TE_1). Figure 3.17a shows a schematic of the simulation

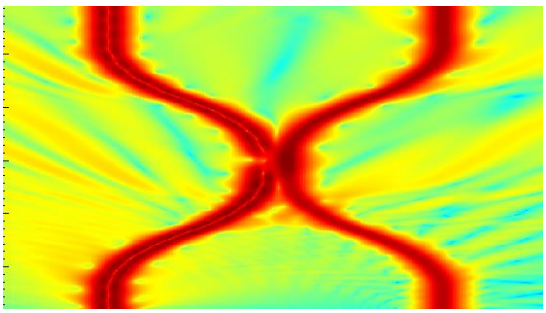
setup in Mode solver to find the required length of the coupler for the interference to occur. The gap between the waveguides is 250 nm as it is found from the previous input coupler study, the coupling length is 3.38 μm, and the power at the end of the coupler is 49.02% for each arm. The interference of the light waves is shown in Figure 3.17b. The power loss in the output coupler is about 1.96%, due to the scattering loss as it is observed in Figure 3.17c. The lost power is due that the bend used (15 μm) introduced a radiation loss especially in the first-order mode waveguide, also due to the leaked light in the coupling region as



(a)



(b)



(c)

Figure 3.17: Output coupler design for athermal MZI. (a) coupler simulation setup. (b) power profile of the output coupler. (c) log power profile.

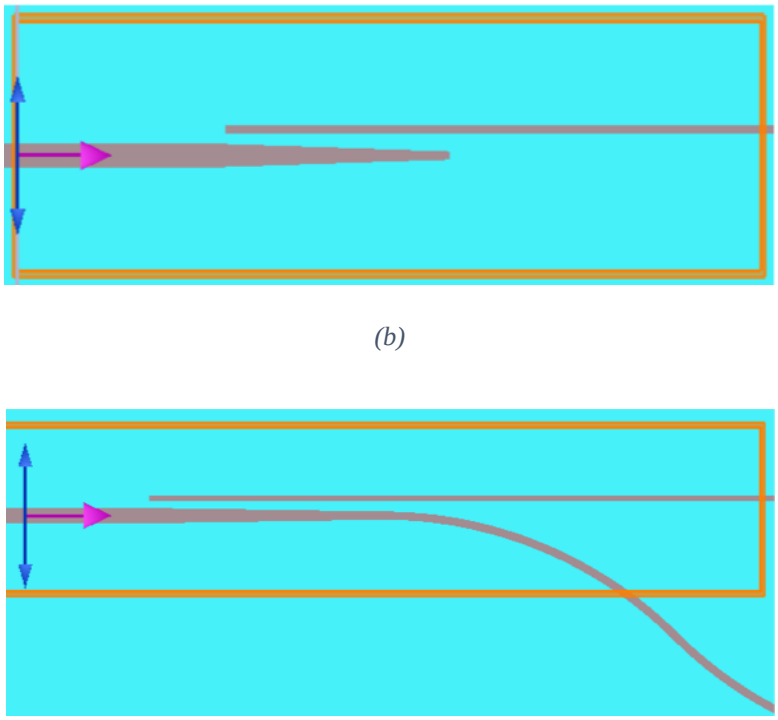
the light is coupled from one waveguide to another. The loss due to the bend can be further minimized if a bigger radius is used, but as described in section 3.3.3, the reduction in power loss for a bigger radius is too small to be considered. Also, a bigger radius means a larger footprint of the design, and this is highly avoided.

3.3.4.3 Other Coupler design

Another coupler design was needed to complete the design of the athermal, and it is due to the limitation of GF-45SPCLO technology. As the optical input and output pads used in this technology is made only for a fixed waveguide width and only for the fundamental mode (TE_0) is presented, a coupler design was required to couple the light from the first-order mode (TE_1) to the fundamental mode (TE_0) to be able to connect it to the optical output pads. The waveguide with a width of 558 nm is used to carry the first-order mode through it to coupler the light back to the fundamental mode two designs were simulated and studied. Figure 3.18 shows the difference between the two designs. Both the designs use a tapered waveguide coupled to another waveguide with a width smaller than the original waveguide. The difference between the two designs is that the design shown on top in Figure 3.18, the input waveguide ends at the end of the taper while the second design the input waveguide is bent away from the coupler at the end of the taper.

Both the designs were simulated, and the result was compared to observe which of the designs will achieve the highest percentage of power transferred from the first-order mode to the fundamental mode. The first design (coupler without the bend) achieved a power transfer of 97.1% at a tapering width of 455 nm and a length of 5.4 μm. Figure 3.19a shows the study

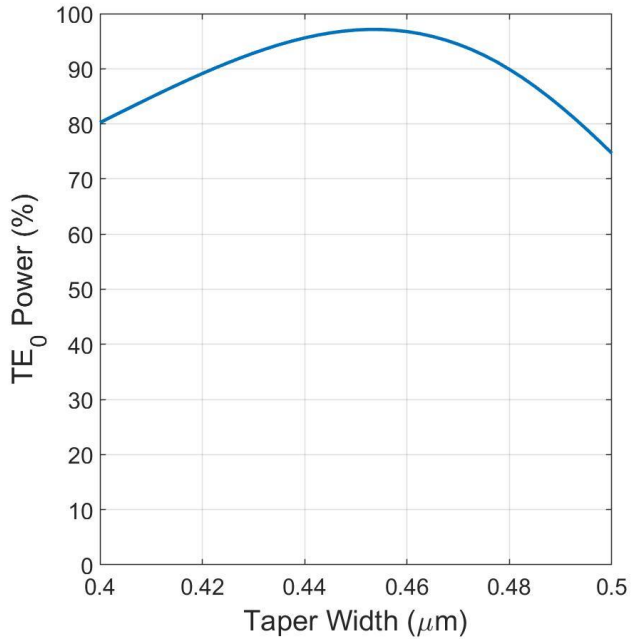
of the taper width and how it affects the power transferred. While the second design (coupler with the bend) achieved a power transfer of 99.8% from the first-order mode to the fundamental mode at a tapering width of 455 nm and a taper length of 8.5 μm. Figure 3.19b shows the percentage of power transferred as a function of taper length.



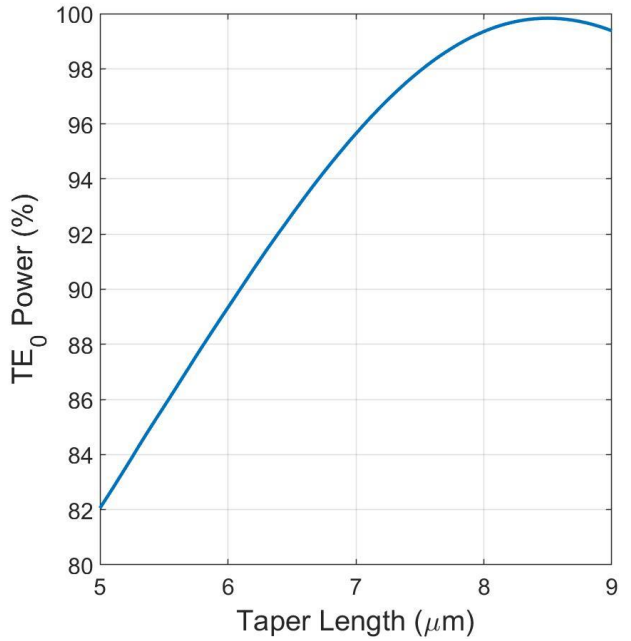
(b)

(a)

Figure 3.18: Coupler design for the output optical pads. (a) coupler without bend. (b) coupler with bend.


100
90
80
70
60
50
40
30
20
10
0
TE0 Power (%)
0.4
0.42
0.44
0.46
0.48
0.5
Taper Width (μm)

(a)


100
98
96
94
92
90
88
86
84
82
80
TE0 Power (%)
5
6
7
8
9
Taper Length (μm)

(b)

Figure 3.19: Study of taper width and length effect on the power transmitted (a) Percentage of TE_0 power transmitted as a function of taper width (b) Percentage of TE_0 power transmitted as a function of taper length.

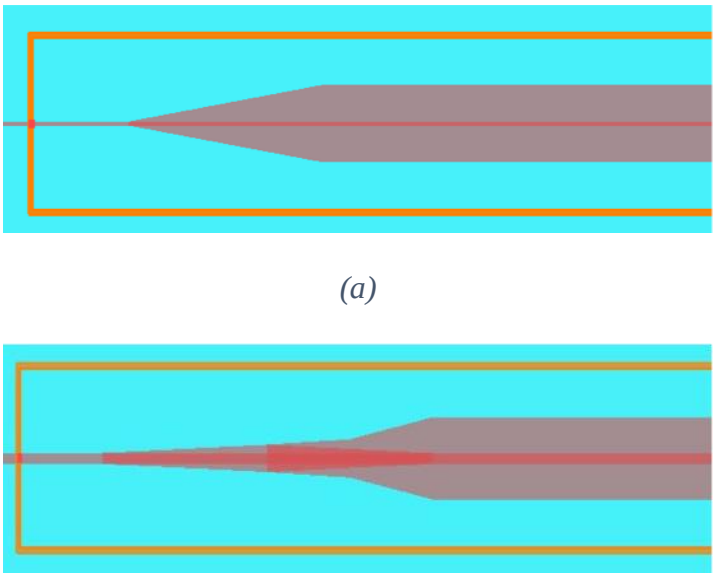
3.3.5 Taper

"The function of the taper is to change the size and the shape of the optical mode to achieve a high coupling efficiency between two waveguides with different cross-sections" [88]. In the design of the athermal MZI more than one taper is required to be designed. This taper design is required due to the constraints by GF-45SPCLO and GF-90WG. For example, the grating couplers that connect the input to the optical fiber light source (laser) is designed for 350 nm waveguide width. The athermal design has a different waveguide width: thus, a taper is required to be designed to change the waveguide width from the designed width to 350 nm and connect it the optical fiber. Another important taper to be designed is the taper to change from strip to rib waveguide as if the change happens without a taper. The light wave will suffer from a great loss due to back reflection and scattering loss because of the mode mismatch.

3.3.5.1 Strip to slab waveguide taper

From the athermal MZI design, the slab waveguide widths w_1, w_3 are 558 and 203 nm, respectively. That means it is required to design two different tapers to connect to each waveguide. Two different approaches were studied to find a good taper design. The first approach is to create a simple linear taper Figure 3.20a, where the slab of the waveguide will start to increase linearly with the length. The second approach (adiabatic approach) is different where the core waveguide starts to increase in width with length, then decreases, and the slab

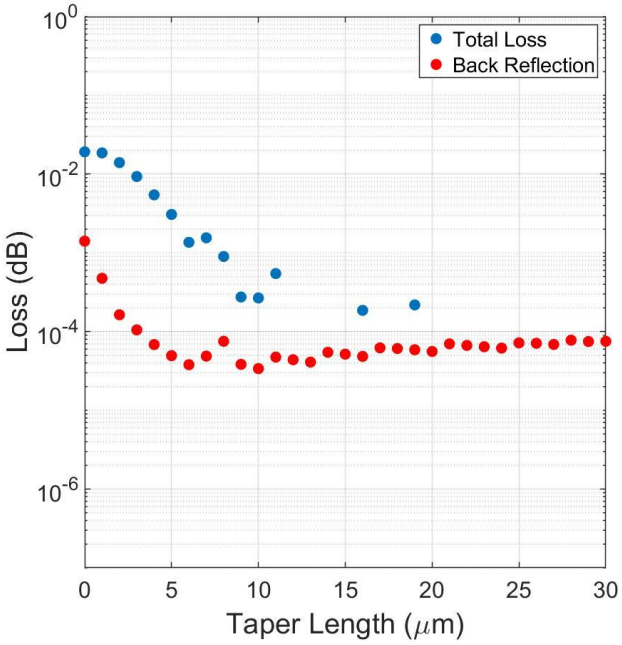
starts to appear and increase slightly for the first part, then an abrupt increase of the slab to the required length Figure 3.20. The adiabatic approach shows a lower back reflection and scattering losses when compared to the linear approach, as in Figure 3.21a as a function of taper length. The taper length was set at 20 μm as it is shown from Figure 3.21b, that the back reflection and scattering losses do not change with length after a certain length.



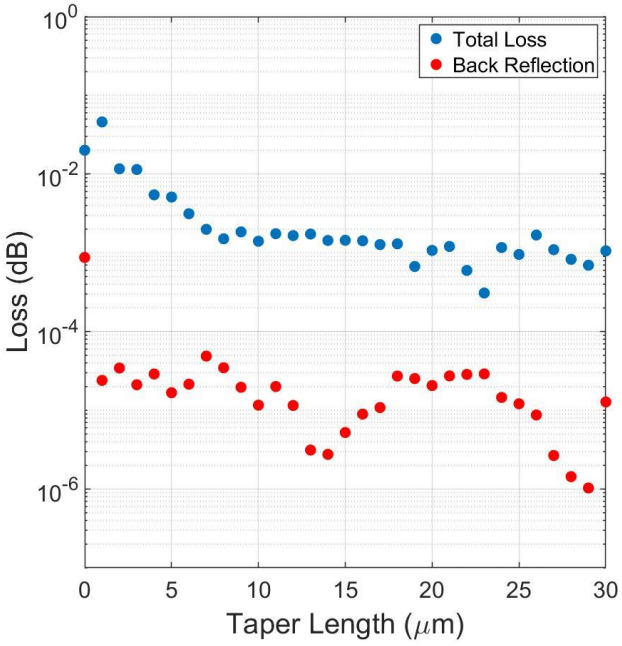
(a)

(b)

Figure 3.20: Proposed strip to slab waveguide design (a) linear design, (b) adiabatic design.



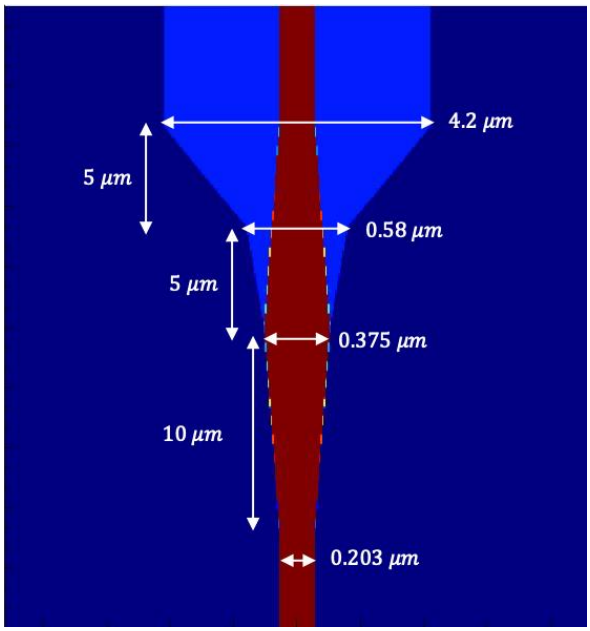
(a)



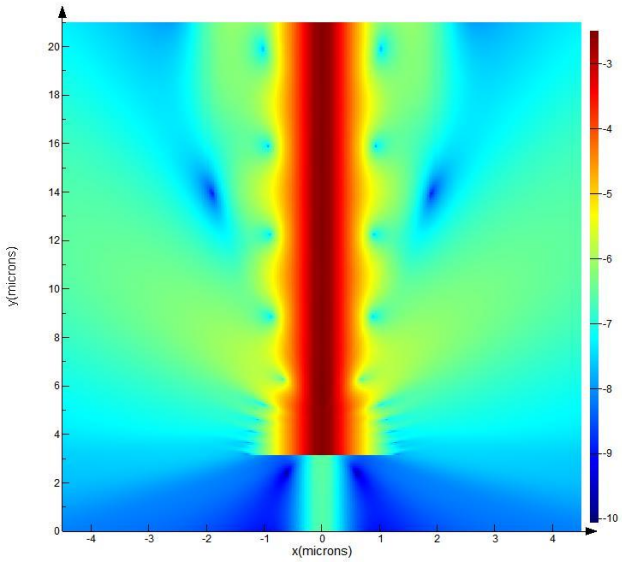
(b)

Figure 3.21: Study of power loss for the two proposed tapers (a) linear design, (b) adiabatic design.

Two designs were required to be optimized using the adiabatic slab taper as the two arms of the MZI has different widths. The same approach was used to optimized both the taper designs. The only difference in optimizing the designs is that for the waveguide width of 203 nm it is optimized for the fundamental mode (TE_0) as this is the mode that the light will propagate in this arm. In comparison, the other waveguide width of 558 nm the design was optimized for the first-order mode (TE_1) as this is the mode that the light will be traveling in. Optimizing the taper design can be different for each mode, for example, optimizing for the same waveguide width will lead to different taper designs each is optimized to minimize the loss for a specific mode. Figure 3.22a shows the optimized design for the waveguide width of 203 nm for the fundamental mode (TE_0) alongside with the power log profile Figure 3.22b. This design offers an insertion loss of 0.0055 dB and a back-reflection loss of $7.8*10^{-5}$ dB. Figure 3.23a shows the optimized design for the waveguide width of 558 nm for the first order mode (TE_1) alongside with the power log profile Figure 3.23b. This design offers an insertion loss of 0.0063 dB and a back-reflection loss of $1.8*10^{-6}$ dB.

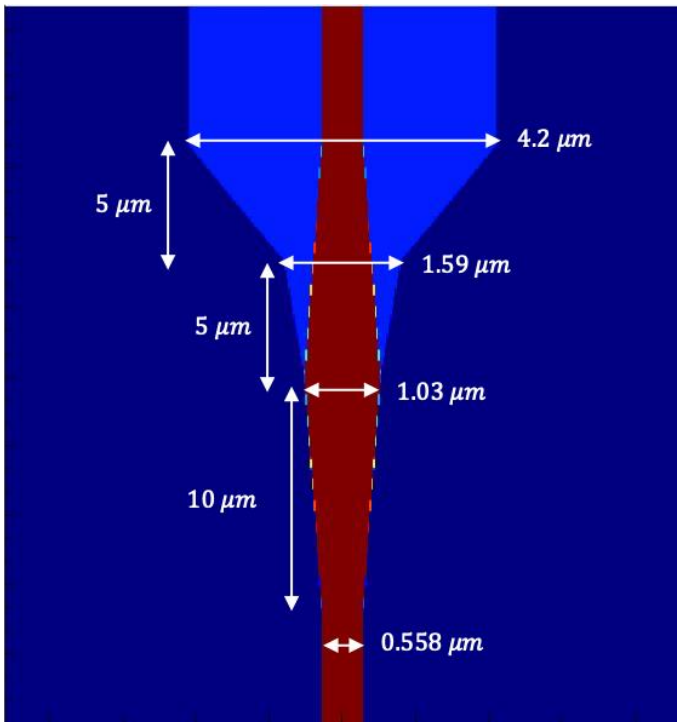


(a)

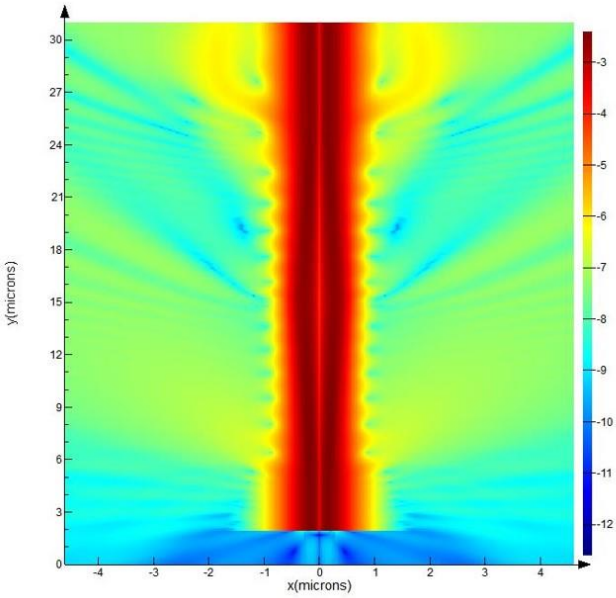


(b)

Figure 3.22: Optimized taper design for a waveguide width of 203 nm, (a) Schematic of the optimized design, (b) Power log profile of the design.



(a)



(b)

Figure 3.23: Optimized taper design for a waveguide width of 558 nm, (a) Schematic of the optimized design, (b) Power log profile of the design.

3.3.5.2 Waveguide taper between two different waveguide widths

Another taper design was required to be simulated and optimized as in the athermal MZI design. Different waveguide width is available in one of the arms (waveguide width changes from 203 nm to 400 nm or vise-versa). A simple linear taper design was used to change the waveguide width with minimum taper length to reduce the footprint of the overall design. Simulating the taper behavior with different taper lengths was conducted, and it was shown that after a certain length of the taper, the scattering loss is fixed. A taper length of 5 μm was used for all the linear taper designs that change between the waveguide widths.

3.4 GDS layout and tape-out considerations

Graphic Data Stream (GDS) is used by GF to fabricate the proposed athermal MZI design. Before creating the GDS, there are several considerations needed to take care-of. For example, the minimum gap between two waveguides (Silicon core waveguide) is set to a certain value; also, the gap between two slabs of two waveguides has another constraint. All of those constraints and others were taken into consideration in the design process of the athermal MZI. The constrains different from one technology stack to another as it is due to the fabrication process.

Figure 3.24 shows the GDS layout indicating all the simulations done for the design. Each part in the GDS layout highlighted by a square is done separately by a different simulation to get the design. The GDS layout can be divided into three main sections, the first section is the input section denoted by the letter A, and it is highlighted by double dashed square. In this section, the input light injected is propagating in the fundamental mode (TE_0) in the lower arm and the mode couples to the first-order mode (TE_1) in the top arm. The second section is the propagation section denoted by the letter B, and it is highlighted by double dashed square. In this section, the light propagates in both arms with different modes (TE_0 in the lower arm and TE_1 in the upper arm). The third section is the output section denoted by the letter C, and it is highlighted by a double dashed square. In this section, both the lights propagating in each arm

will be brought closer and interfere with each other; the waveguides will be separated to be connected to the output.

In section A, there are three main sub-designed done to complete this section. The designs are marked 1,2 and 3 in

Figure 3.24. The first design (noted by 1) is the design of the input coupler, which was described in section 3.3.4.1. The input coupler will couple the light from the input waveguide (TE_0) to the second waveguide (TE_1) with achieved power percentage of 47.4% in each waveguide (TE_0 in the first waveguide and TE_1 in the second waveguide), only 2.74% will be coupled as TE_0 in the second waveguide. It can be ignored as it is relatively small. The length of the coupling region is 855 nm, with a separation gap of 250 nm between the two waveguides. The second design is the strip to slab taper design, which was discussed in section 3.3.5.1. Using the proposed adiabatic taper, the back-reflection and scattering losses are minimized for a taper length of 20 μm. The third design (noted by 3) is the bend radius used in the design, which was discussed in section 3.3.3, the bend radius used in all the designs of the athermal MZI (in all of the sections of the design A, B, and C) is 15 μm. The three designs combined to form the input section (A).

In section B, where mainly the light is propagating through the arms, only the design of the taper is used to change from a waveguide width of 203 nm to a waveguide of 400 nm. The linear taper of a length of 5 μm is used for the taper. The bends shown in this section uses a radius of 15 μm. This section is the core of the athermal MZI design as due to the difference of the waveguide width and the selected arm lengths shown in Table 3.2 the device can be thermally insensitive to temperature as the optical path difference does not change with the designed temperature range.

In section C, which is the output section of the athermal MZI, where the interference occurs of the two lights coming from each waveguide, it consists of mainly three sub-designs (noted by 2,5 and 6). The first design (noted by 2) is the strip to slab taper used in the input section A and explained in section 3.3.5.1. The second design (noted by 5) is the output coupler design explained in section 3.3.4.2, where the light is coming from both the waveguides and

coupler will allow the lights to interfere with each other then it will be coupled again to have 49.02% of the interfered light in each waveguide the remaining power is lost as scattering loss. The output coupler length is 3.38 μm, with a gap of 250 nm. The third design (noted by 6) is the coupler design to be able to connect the arm of the athermal MZI to the optical pads due to the limitation of Gf-45SPCLO technology. This design is described in section 3.3.4.2, the coupler length used in this design is 8.5 μm with a taper that change the waveguide width from 558 nm to 455 nm with a power transfer of 99.8% to the fundamental mode which then connected to the optical pads.

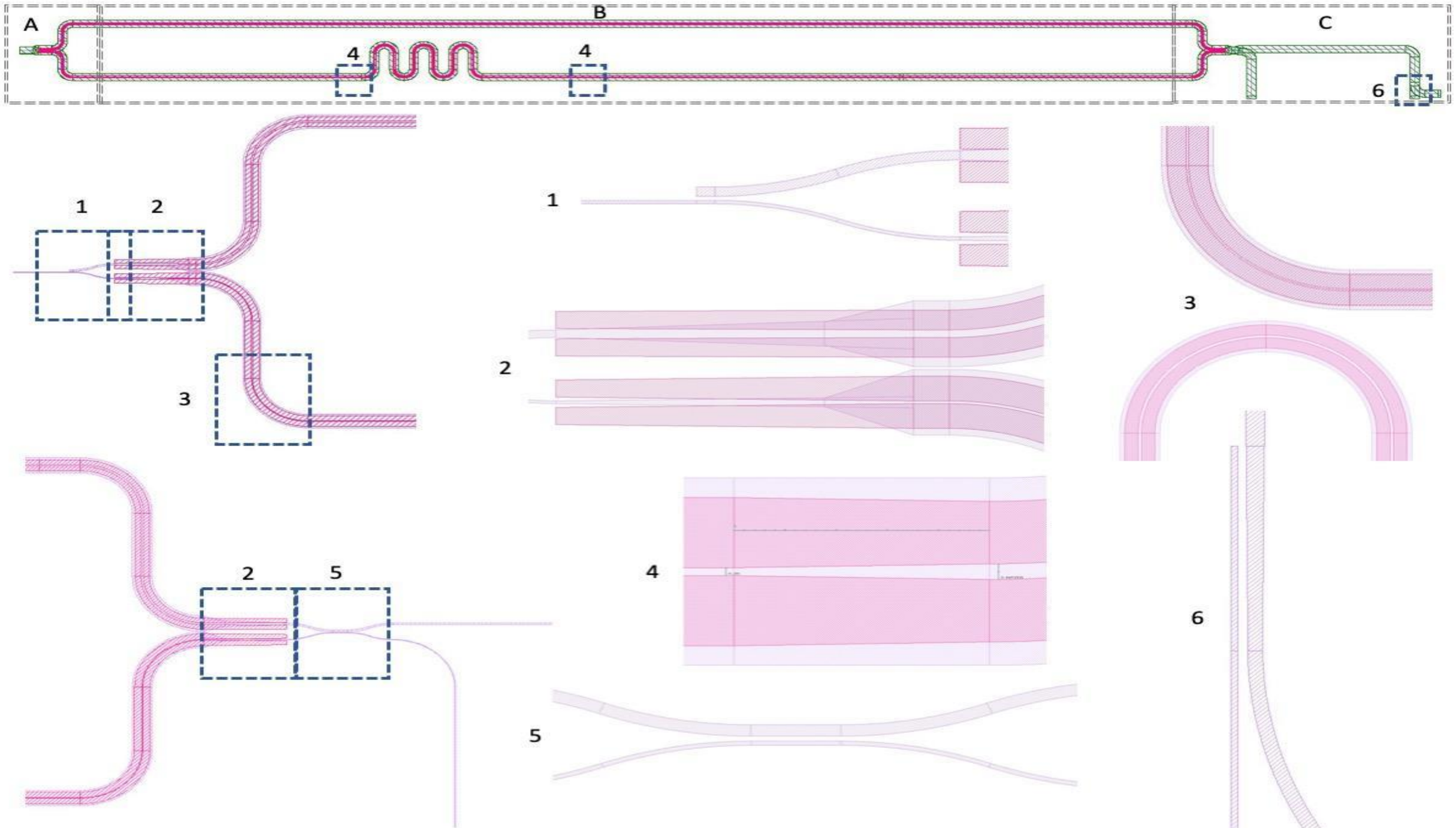


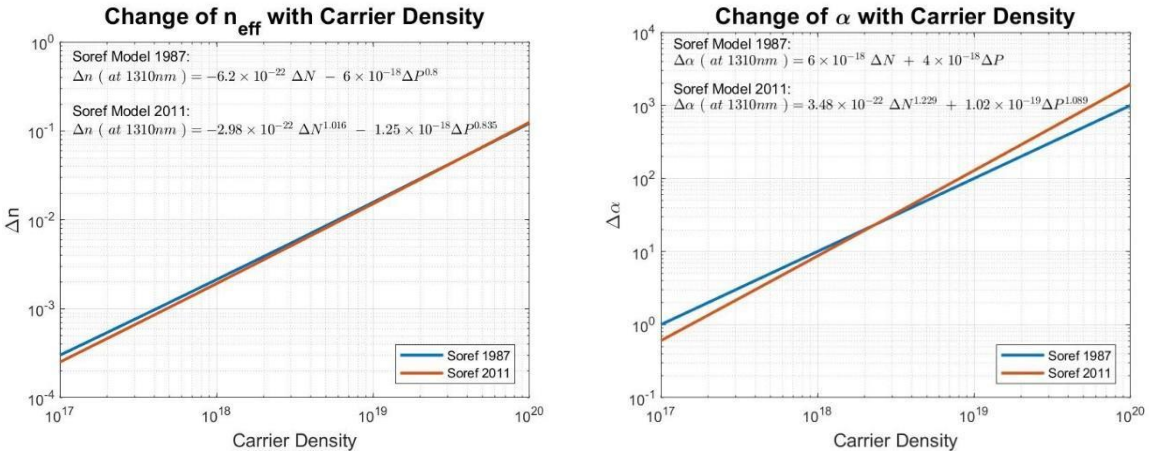
Figure 3.24: 45SPCLO GDS layout for athermal MZI design.

Chapter 4: Modulator design – MZM

4.1 Free carrier dispersion effect (Soref Models)

On a waveguide level, the optical modulation of a signal is based on the Free-Carrier Plasma Dispersion Effect, which consists of two parts: optical part (optical mode) and the electrical part (free carrier distribution). The interaction between these two parts leads to the change in the refractive index of the material, and it was studied by Soref [3], [4]. Empirical equations were provided by the study, as shown in section 2.4.2. The effect was studied in two different papers by Soref, one in 1987, and a newer one in 2011.

The difference in the result obtained from the two papers is shown in Figure 4.1. The new proposed model by Soref in 2011 was used in the calculation of the refractive index change and the absorption coefficient change. The complete simulation procedure is shown in Figure 4.2. It consists of creating the PN junction design by doping the silicon waveguide with implants for n-type and p-type, then use the doping profiles to calculate the charge density as a function of the voltage applied across the PN-junction in reversed bias. The charge distribution is used in Mode solver to solve for the change of refractive index of silicon using the 2011 Soref model. This procedure is repeated for each voltage point applied on the junction as different voltages will lead to a different amount of change in the refractive index.



(a) *(b)*

Figure 4.1: Comparing the Soref models for free carrier dispersion effect (a) change of refractive index with carrier density (b) change of absorption coefficient with carrier density.

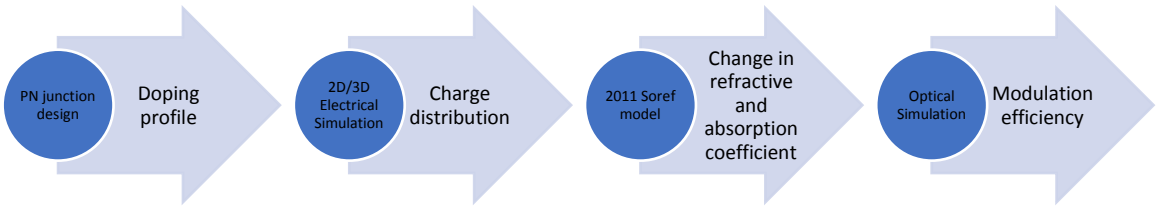


Figure 4.2: Free carrier effect simulation flow chart.

Figure 4.1 shows the difference between the two models proposed by Soref. The difference of the first model, which was reported in 1987 [3] and the model reported in 2011

[4], is summarized in Table 4.1. The observation can be categorized into two different regions; the first region is where the carrier concentration is low (lower than $5*10^{18}$ cm^{-3}) and the second region where the carrier concentration is high (higher than $5*10^{18}$ cm^{-3}). In the low concentration region, the change of the refractive index is lower in the 2011 model than the 1987 model, while it is almost the same for the high concentration region. Meanwhile, the change in the absorption coefficient in the low concentration region 2011 model is lower than the 1987 model, while in the high concentration region 2011 model has a bigger change in the absorption coefficient than the 1987 model. This observation is important as it is the main factor affecting the modulation efficiency and loss in the modulator. Also, from the equations of the two models shown below, it is clear that it is preferable to have the p-type with a higher concentration than the n-type due to that it will lead to a higher effect in the refractive index and lower effect on the absorption coefficient. Thus, a higher modulation efficiency with lower losses is achieved. The empirical equations for each model are shown below:

- 1987 Model [3]:
  - $\Delta\alpha = \Delta\alpha_e + \Delta\alpha_h = 6 * 10^{-18}\,\Delta N_e + 4 * 10^{-18}\Delta N_h$
  - $\Delta n = \Delta n_e + \Delta n_h = -6.2 * 10^{-22}\,\Delta N_e - 6 * 10^{-18}\Delta N_h$
- 2011 Model [4]:
  - $\Delta\alpha = \Delta\alpha_e + \Delta\alpha_h = 2.48 * 10^{-22}\,\Delta N_e^{1.229} + 1.02 * 10^{-19}\Delta N_h^{1.089}$
  - $\Delta n = \Delta n_e + \Delta n_h = -2.98 * 10^{-22}\,\Delta N_e^{1.016} - 1.25 * 10^{-18}\Delta N_h^{0.835}$

Table 4.1: Difference between 1987 and 2011 Soref models for free carrier dispersion effect

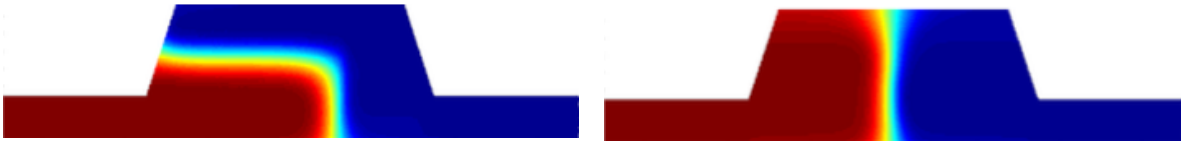
Soref Models		1987 Model	2011 Model
Low concentration carrier density	Δn	Higher	Lower
	Δα	Higher	Lower
High concentration carrier density	Δn	Same	Same
	Δα	Lower	Higher

4.2 Doping profiles and their modeling

Doping profile is one of the main factors that affect modulation efficiency as its directly proportional to the free carrier dispersion effect, which related to the refractive index as it is described in section 4.1. Different doping profiles will lead to a different change in the refractive index; thus, the phase change will be different at the same time it will affect the absorption coefficient. An ideal doping profile is where it maximizes the change of refractive index with applied voltage and minimize the absorption coefficient with voltage. In this section, different doping profiles were studied with different configurations and the amount of dopant concentrations. The doping profile configuration can be expressed as a linear PN-junction doping profile and an L-shaped PN-junction profile. The L-shaped PN-junction profile was used by GF, which is provided to be studied as a part of the project collaborations while the linear PN-junction profiles are proposed by this thesis. All the profiles were simulated and compared. Figure 4.3a shows a schematic of GF L-shaped doping, while the linear doping schematic is shown in Figure 4.3b. In order to simulate the doping profile and get the result as

close as to the real value as possible, a small study of the sidewall angle was conducted before the study of doping profiles. Figure 4.4 shows three different sidewall angles studied for this waveguide with an angle of 71 degrees. This angle is used as a part of the 45SPCLO technology limitation. As it is observed from the Figure 4.4, the first study is the study with an ideal angle of 90°; the second study is where the design waveguide width of 350 nm is measured at the center. Finally, the last study is where the design waveguide width of 350 nm is measured at the base

of the waveguide.



(a) *(b)*

Figure 4.3: Doping profile schematic (a) L-shaped doping (b) Linear doping.



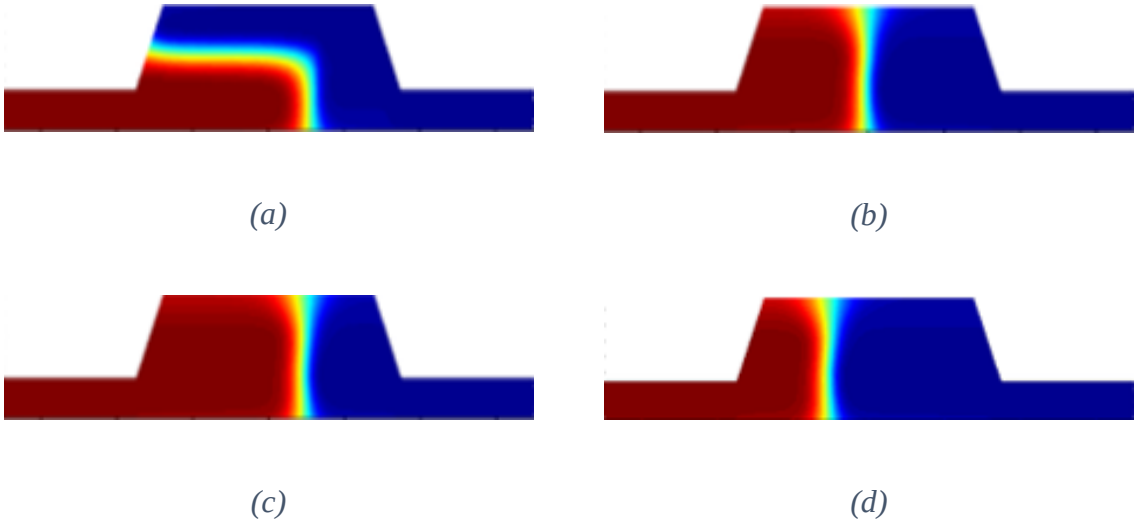
(a) *(b)* *(c)*

Figure 4.4: Side wall angle study (a) ideal angle of 90o (b) waveguide width measured at the middle of waveguide (c) waveguide with measured at the base of the waveguide.

Seven different doping profile was studied in order to optimize the doping profile and minimize the value of $V_\pi L$ alongside with the absorption coefficient; thus, the power loss will be reduced, and high modulation efficiency is achieved. The seven different cases studied are as follows:

- Case 1: GF default doping profile (L-Shaped doping).
- Case 2: GF default doping profile (Linear doping).
- Case 3: GF default doping profile (Linear doping with an offset of +50 nm).
- Case 4: GF default doping profile (Linear doping with an offset of -50 nm).
- Case 5: Proposed low dose doping profile (Linear doping with no offset).
- Case 6: Proposed low dose doping profile (Linear doping with an offset of +50 nm).
- Case 7: Proposed low dose doping profile (Linear doping with an offset of -50 nm).

In Case 1, the doping profile available in 45SPCLO technology was studied without any change in the dose values of the location of the PN junction, while in cases 2 – 4 some of the layers of the original doping available in 45SPCLO was removed/altered to create a doping profile with linear PN junction. The difference between cases 2 – 4 is that the location of the PN junction is either in the middle of the waveguide (case 2), 50 nm to the right side of the center of the waveguide (case 3) or 50 nm to the left of the center of the waveguide (case 4). The remaining cases 5 – 7 use a low dose linear doping profile where the PN junction is either in the center of the waveguide (case 5), 50 nm right of the center of the waveguide (case 6) or 50 nm to the left of the center of the waveguide (case 7). Figure 4.5 shows some of the cases to observe the difference between them. Table 4.2 shows the values of the doping range used in each case.



(a) *(b)*

(c) *(d)*

Figure 4.5: Doping profile schematic for the different cases studied (a) L-shaped doping (b) Linear doping with the PN junction in the middle of the waveguide (c) Linear doping with the PN junction 50 nm to the right of the center of the waveguide (d) Linear doping with the PN junction 50 nm to the left of the center of the waveguide.

Table 4.2: Doping profile cases with their parameters used in the study.

Case Number	n-type dose (cm-3)	p-type dose (cm-3)	Offset from the center of the waveguide (nm)
1	$>10^{17}$	$>10^{18}$	L-Shaped PN junction
2	$>10^{17}$	$>10^{18}$	0
3	$>10^{17}$	$>10^{18}$	+50
4	$>10^{17}$	$>10^{18}$	-50

5	$>10^{17}$	$<10^{18}$	0
6	$>10^{17}$	$<10^{18}$	+50
7	$>10^{17}$	$<10^{18}$	-50

4.2.1 SRIM simulation

SRIM simulation, which stands for The Stopping and Range of Ions in Matter simulation Figure 4.6 shows an overview of the simulator. This simulation was important to be done due to the limitation of the doping information provided by GF. The doping information provided by GF was the dopant species, the energy of implantation, and the dose in cm^{-2}. In order to have a full understanding of the doping profile, a SRIM simulation was conducted. 21 different simulation was conducted for 14 different mask layers. The reason behind the different simulations is that for each mask layer and each species and implantation energy used, a different doping behavior will occur. The dopant behavior can be described by the ion range, peak concentration, straggle, skewness, and kurtosis of the implants. Figure 4.7

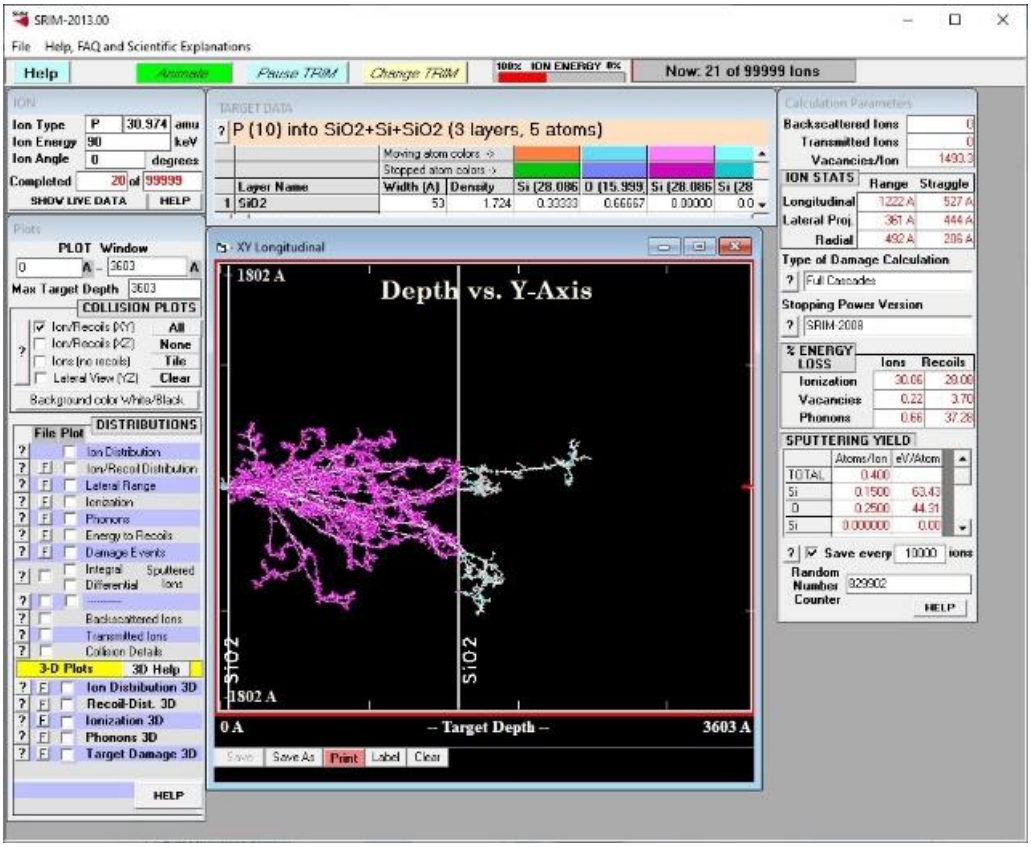
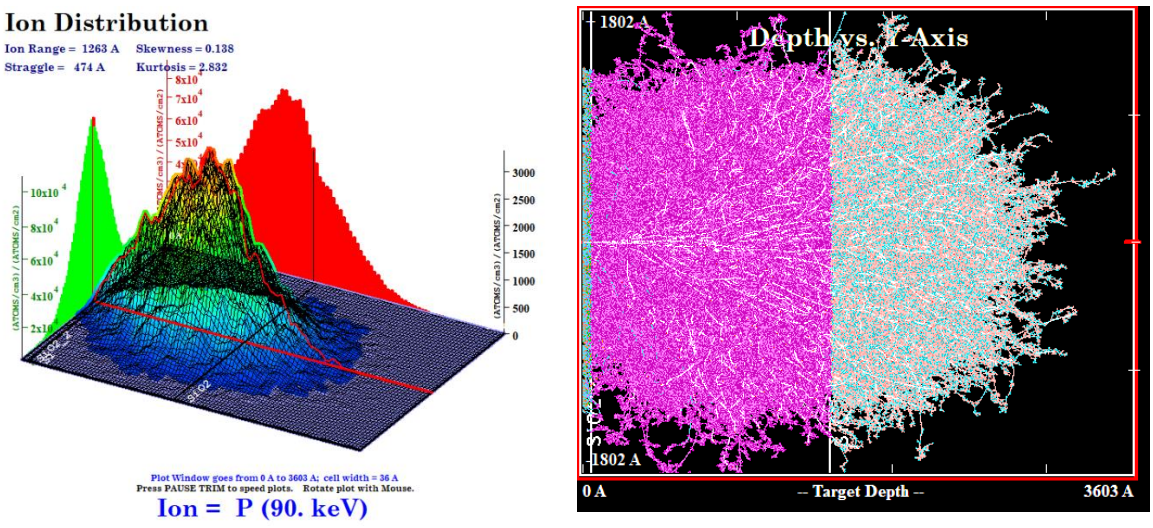


Figure 4.6: SRIM software overview.



(a) *(b)*

Figure 4.7: Example of SRIM simulation for one mask layer (a) 3D simulation of implanting phosphorous in Silicon (2) 2D simulation of implanting phosphorous in Silicon.

4.3 Signal modulation response

This section was studied with help from one of my colleagues (Marios Papadovasilakis), The study, which is represented in this section, was done together, and we run different simulations and compare them in order to optimize the signal response of the modulator.

The response of the signal is affected by the doping profile, as the doping profile (PN-junction shape and free carrier concentration) is directly proportional to the resistance and capacitance of the junction. The resistance of the junction can be given by [52]:

$$R_j[\Omega.m] = \left(\frac{w}{2} + y_p\right) R_{srp} + \left(\frac{w}{2} - y_n\right) R_{srn} - \left(\frac{w}{2} + y_{p++}\right) R_{ssp} + \left(y_{n++} - \frac{w}{2}\right) R_{ssn}$$

where $R_{srn}, R_{srp}, R_{ssn}\ and\ R_{ssp}$ are the sheet resistance of the n-doped rib, p-doped rib, n-doped slab, and p-doped slab, respectively. $y_p, y_n, y_{p++}\ and\ y_{n++}$ are the distance from the middle of the PN junction to the start of the p-doped region, distance from the middle of the PN junction to the start of the n-doped region, distance from the middle of the PN junction to the start of the highly p-doped region and distance from the middle of the PN junction to the start of the highly n-doped region, respectively. Observing the resistance equation, it is clear that the resistance is changing with applied voltage due to change in the depletion region with voltage. **Error! Reference source not found.**shows an illustration of the PN-junction and how $y_p, y_n, y_{p++}\ and\ y_{n++}$ are defined. Using Lumerical FEM solver (Lumerical device software package) sheet resistance can be extracted in $\Omega.cm$. The total sheet resistance in the software can be given by:

$$R_{total} = R_n + R_p$$

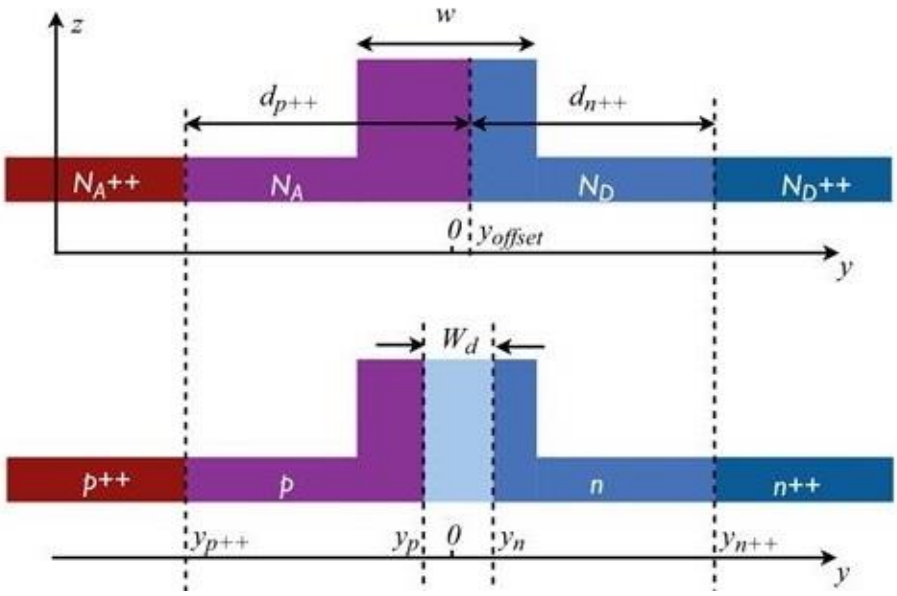


Figure 4.8: PN Junction illustration [52].

Where $R_n and\ R_p$ are the sheet resistance of the n-doped region and p-doped region, respectively.

The capacitance of the junction is also an important factor to look at to get a comprehensive response study of the PN-junction. The capacitance of the junction for each case is given by [52]:

$$C_j[^F/_m] = t_{rib}\sqrt{\frac{q\epsilon_0\epsilon_s}{2\left(\frac{1}{N_D}+\frac{1}{N_A}\right)(V_{bi}-V)}}$$

Where t_{rib} is the slab thickness, ϵ_s is the relative permeability of the material (Silicon), V_{bi} is the built-in protentional, V is the applied voltage, q is the electron charge constant, and N_D, N_A is the dopant concentration and acceptor concentration, respectively. In the doping profile simulation, the capacitance as a function of voltage for each case studied is extracted and shown

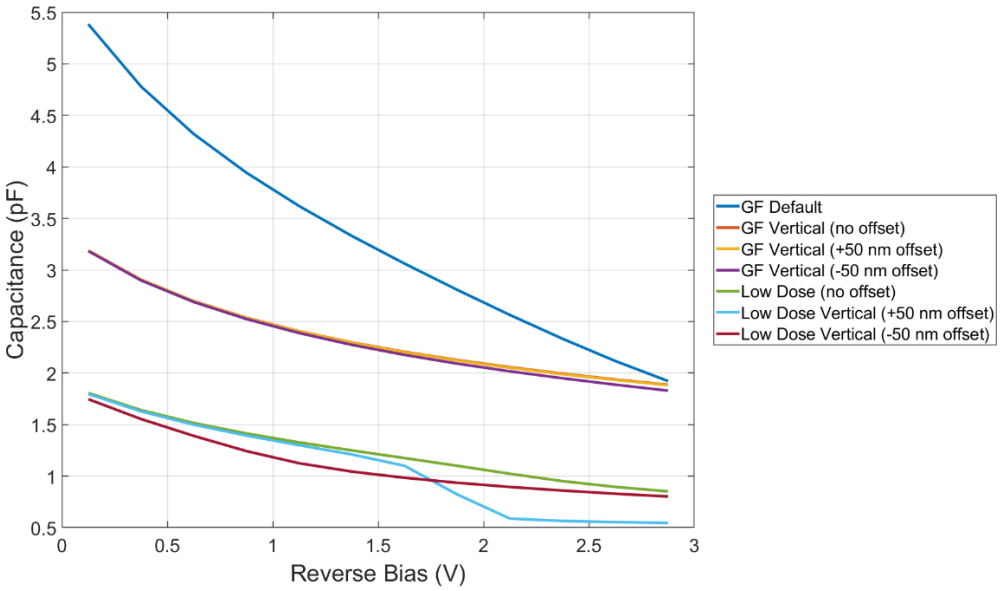


Figure 4.9: PN junction capacitance as a function of voltage (reversed biased) for the different study cases of doping.

in Figure 4.9. The capacitance is reversely proportional to the voltage applied across the junction (reverse biased), as the applied voltage increase the capacitance decreases.

The cutoff frequency (3dB) due to the RC time constant is given by [52]:

$$f_{RC} = \frac{1}{2\pi R_j C_j}$$

The cutoff frequency is the limit of frequency of what is achieved for the doping profile due to the resistance and capacitance of the profile. This equation is used to determine the speed limit of the modulation in the athermal MZM design, which is the athermal MZI design explained earlier with the optimal doping profile chosen from the case study. On the other hand, for the RR modulator the cutoff frequency (f_c)is different, due to the photon lifetime in the coupled ring, and it is given by [52]:

$$\frac{1}{f_c^2} = \frac{1}{f_{RC}^2} + \frac{1}{f_{\tau_p}^2}$$

$$f_{\tau_p} = \frac{1}{2\pi\tau_p}$$

Where τ_p is related to the total quality factor, Q_t which is given by:

$$\tau_p = \frac{Q_t}{\omega_0}$$

$$\frac{1}{Q_t} = \frac{1}{Q_c} + \frac{1}{Q_i}$$

$$Q_i = \frac{2\pi n_g}{\lambda\alpha}$$

$$Q_c = -\frac{\pi L_{rt} n_g}{\lambda \log_e|t|}$$

Q_i is the intrinsic quality factor, Q_c is the coupling determined by the quality factor. The intrinsic quality factor is the same for all the cases as it depends only on the group index, wavelength and absorption factor, and all those parameters are fixed for all the case, due to the same waveguide design used. Figure 4.10 shows Q_c as a function of power coupling coefficient for different cases studied. The total cutoff frequency, which is determined by both the photon lifetime and the RC time constant, is changing with the voltage applied and differs from a ring radius to another. Three different cases were studied with three different radiuses. As the interest radius is 15 μm, the different case was chosen as follows:

- Case R_1: RR radius of 5 μm (smaller than the radius of interest).
- Case R_2: RR radius of 15 μm (radius of interest).
- Case R_3: RR radius of 25 μm (larger than the radius of interest).

As observed from Figure 4.10, smaller ring radiuses tend to have a higher photon lifetime cutoff frequency, but at the same time, from our previous study, it leads to higher scattering losses. A fully comprehensive study was conducted for the three main cases mentioned above, and the total cutoff frequency was observed for the maximum change of the voltage applied (3.3 V). Figure 4.11 shows the different cases studied, a total of six cases (two cases per radius ring one with zero voltage and another with a maximum voltage applied of 3.3 V).

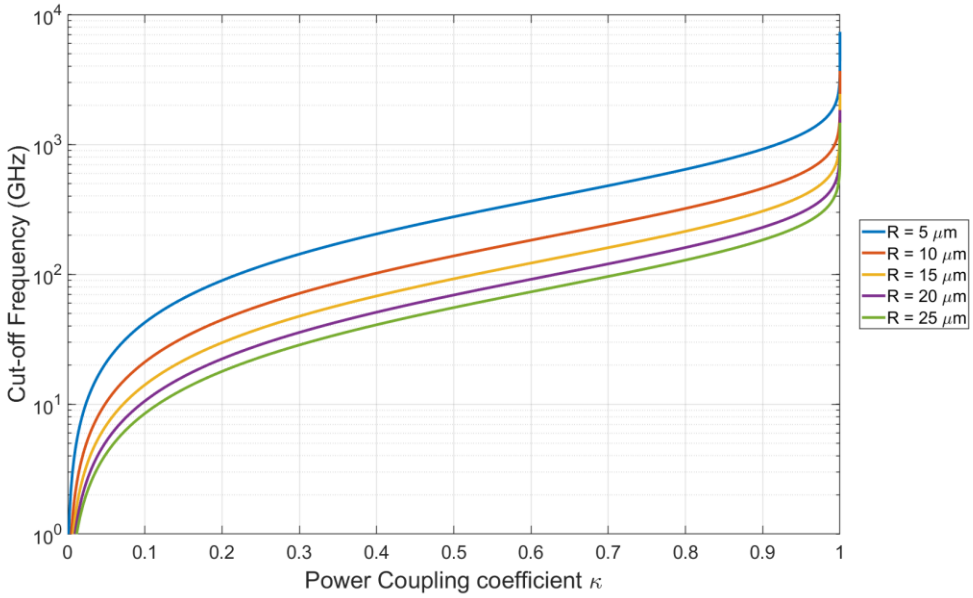
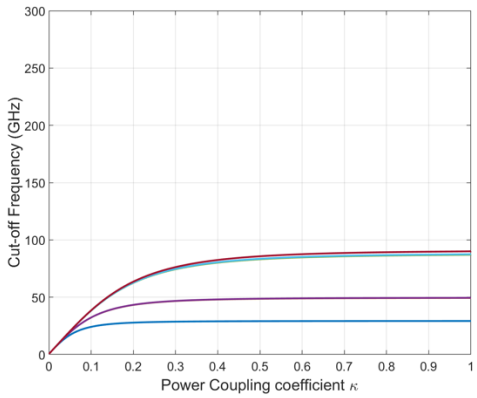
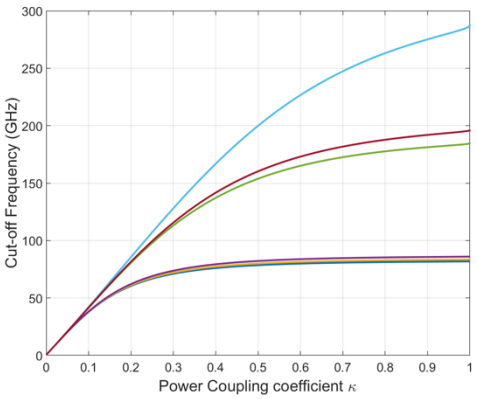


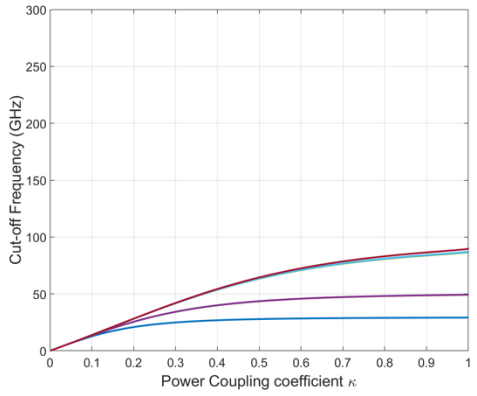
Figure 4.10: Cut-off frequency for different ring radius.



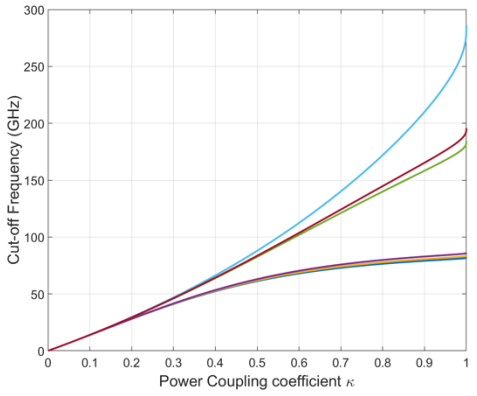
(a)



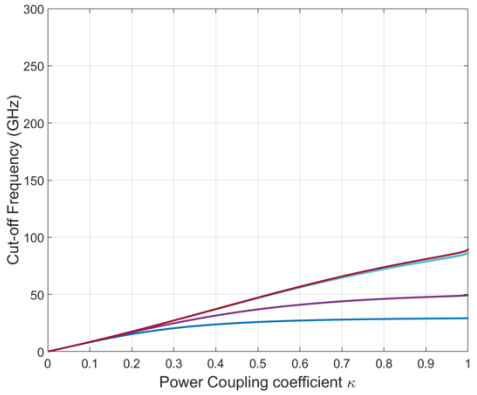
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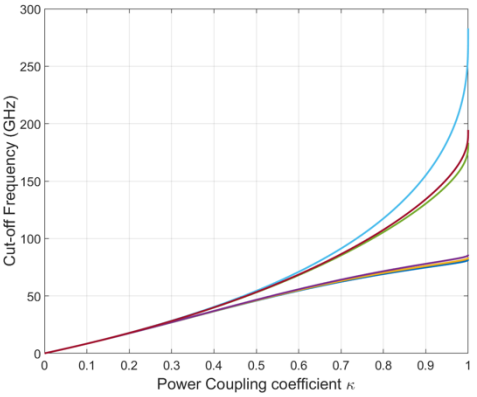
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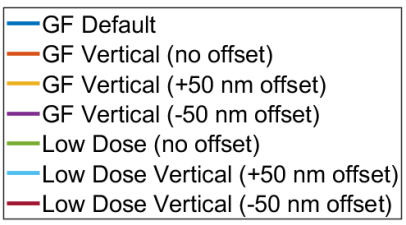
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(f)



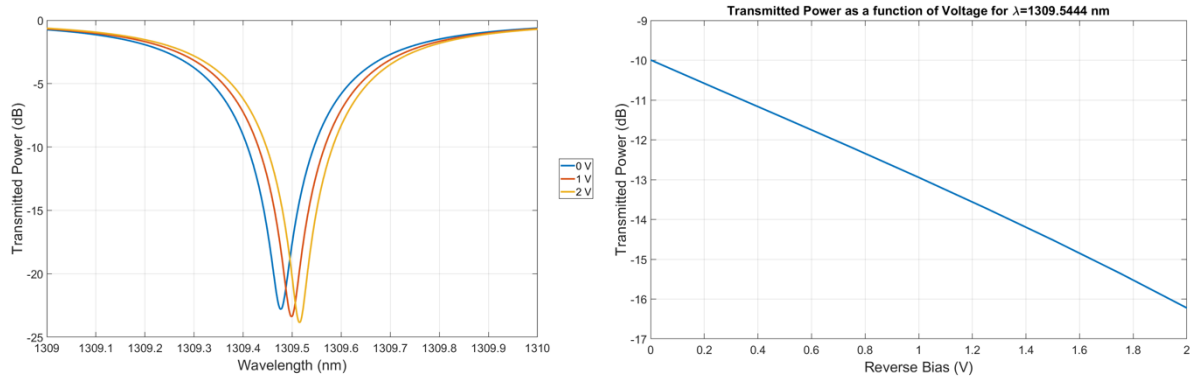
(g)

Figure 4.11: Cut-off frequency study for different doping profiles alongside with different ring radiuses R with applied reversed voltage V (a) R = 5μm, V = 0 V (b) R = 5μm, V = 3 V (c) R = 15μm, V = 0 V (d) R = 15μm, V = 3 V (e) R = 25μm, V = 0 V (f) R = 25μm, V = 3 V (g) plot key showing the colors representing each doping profile.

4.4 Doping profile optimization for extinction ratio and cutoff frequency

The doping profile study shows how the doping is affecting the optical response of the RR. Different doping profile leads to improvements in certain aspect while it is worse its others. For example, the doping study shows that linear doping will lead to a higher cutoff frequency, thus a higher modulation speed when compared to the L-shaped doping profile. Lower doping concentration will lead to lower losses, but at the same time higher $V_\pi L$ values which are not recommended. Comparing all the doping profile cases studied for each ring radius showed a promising result for two sets of parameters. The first set is where the ring radius is 15 μm with a linear doping profile. The second set is where the ring radius is 20 μm with an L-shaped doping profile.

The first set offers the highest speed of modulation when it is compared to the other doping profiles. The cutoff frequency is about 40.1 GHz at zero volts. This cutoff frequency is achieved with a ring radius of 15 μm, a coupling gap between the straight waveguide (which is one of the arms of the MZI) of 200 nm with a straight coupler region of 150 nm. Setting the central wavelength of 1309.54 nm, an extinction ratio of 6.23 dB is achieved. Transmission power for the discussed doping profile is shown in Figure 4.12a as applied voltage changes. To have a clear power transmission of the proposed doping profile, Figure 4.12b is created by taking the value of the power transmitted at the central wavelength as the applied voltage changes. As it is observed from Figure 4.12 at zero volts, the transmission at the central wavelength (frequency) is about -10 dB, and it changes linearly with the applied voltage (reverse biased) till it reaches about -16.23 dB at 2 V. The study was done for peak to peak

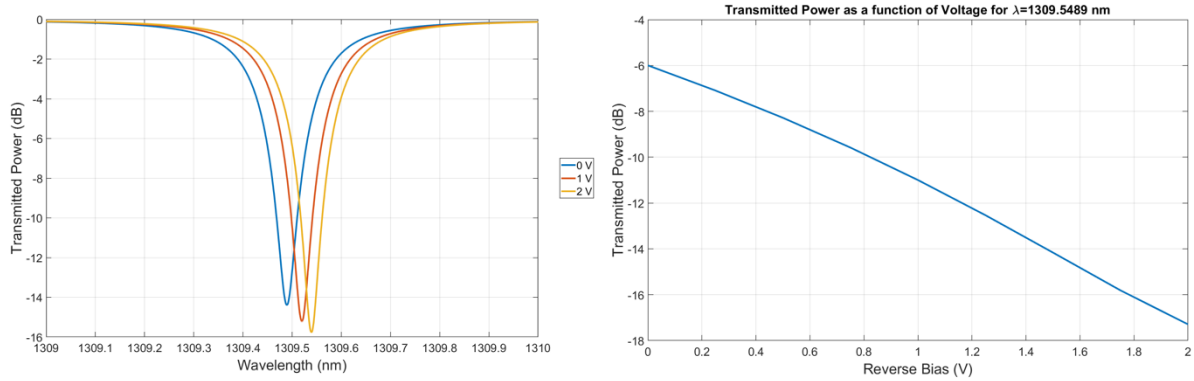


(a) *(b)*

Figure 4.12: Transmission power in dB for the highest modulation speed doping profile (a) Transmission as a function of applied voltage (b) Transmission at the central frequency.

(V_{PP}) of two volts. Thus, it is assumed that the voltage is set at 1 V as DC biased and the RF signal applied to have V_{PP}.

The second set offers the highest ER of 10.495 dB, while the cutoff frequency is only 17.35 GHz. The parameters used to achieve this performance are ring radius of 20 μm,



(a) *(b)*

Figure 4.13: Transmission power in dB for the highest ER doping profile (a) Transmission as a function of applied voltage (b) Transmission at the central frequency.

separation gap between the straight input waveguide and the RR is 280 nm with no straight coupling region (this is different from the above-discussed profile as it has a straight coupling region of 150 nm). The doping profile was studied at the same central wavelength 1309.54 nm, which is the same for the previous profile. Figure 4.13a shows the power transmission with different applied voltages as a

The function of wavelength. Figure 4.13b is the power transmission at the central wavelength as a function of voltage. As it is observed from Figure 4.13, it is clear that this doping profile offers a better ER, but the change in transmission with voltage is not as linear as the previous doping profile. At the same time, the main intent of this thesis is to achieve an ultra-high speed athermal modulator. Thus, the first doping profile (linear doping) is preferred over the L-shaped doping profile.

4.5 Doping profile voltage characteristics

Applying voltage in reversed bias across the PN-junction changes the carrier distribution, thus change the phase and the propagation loss of the electromagnetic wave, which was described in detail in section 4.1. A voltage study is conducted to explore which of the proposed doping profile will minimize $V_{\pi}L$ (which is the voltage needed to have 1 π phase shift in a cm of waveguide length) and minimize the propagation loss. Figure 4.14 shows how the phase-shift experienced in the waveguide is changing with the applied voltage. For the first case (GF default), the $V_{\pi}L$ is 0.49 V.cm, which means it required to apply 0.49 V on the waveguide for 1 cm to achieve a π phase-shift in the electromagnetic wave traveling in the waveguide. Table 4.3 shows the different $V_{\pi}L$ for each case studied. All of the doping profiles

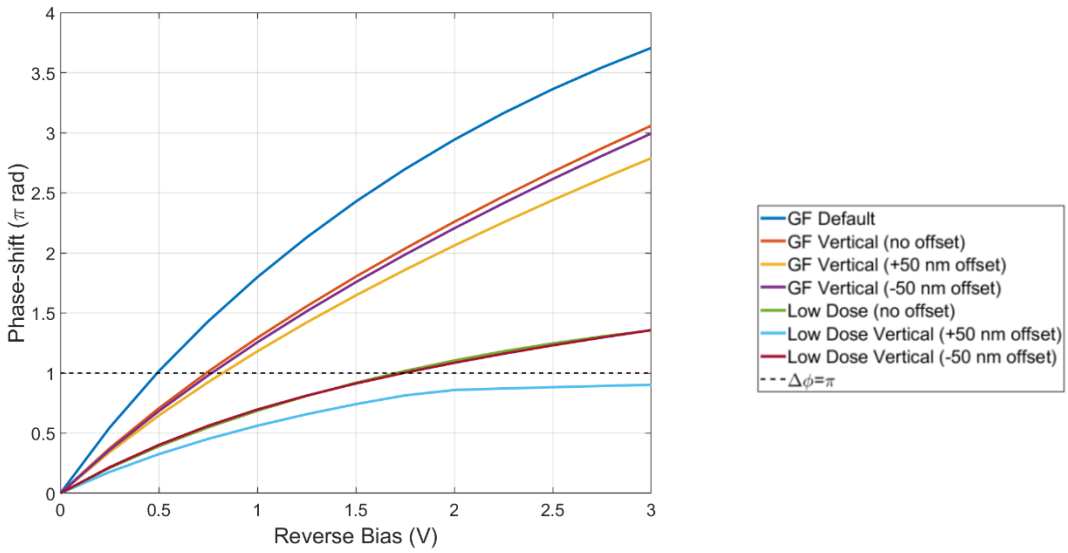


Figure 4.14: Phase-shift as the reverse voltage applied is changed for the different doping profile cases.

achieved a π phase shift within the range of the voltage applied except for case number 6, thus this is the worst doping profile, and it will not be used even if the other studies recommend this profile.

Table 4.3: Comparing $V_\pi L$ for the different cases of doping profiles

Case number	description	$V_\pi L$ (V.cm)
1	GF default	0.49
2	GF Vertical (no offset)	0.74
3	GF Vertical (+50 nm offset)	0.82
4	GF Vertical (-50 nm offset)	0.77
5	Low Dose (no offset)	1.71
6	Low Dose (+50 nm offset)	-
7	Low Dose (-50 nm offset)	1.74

The same study was conducted, but at this time, observing the loss of the propagation power of the electromagnetic wave in the waveguide. The power loss in the propagation tends to get lower as the reverse voltage applied increases. The reduction in propagation losses is because as the applied voltage increases, the carriers are pushed away from the depletion region of the PN junction, and the waveguide becomes more intrinsic. The analysis of the applied voltage effect on the power loss is described in detail in section 4.1. Figure 4.15 shows how the loss is changing as a function of the applied voltage for the different doping profile cases studied.

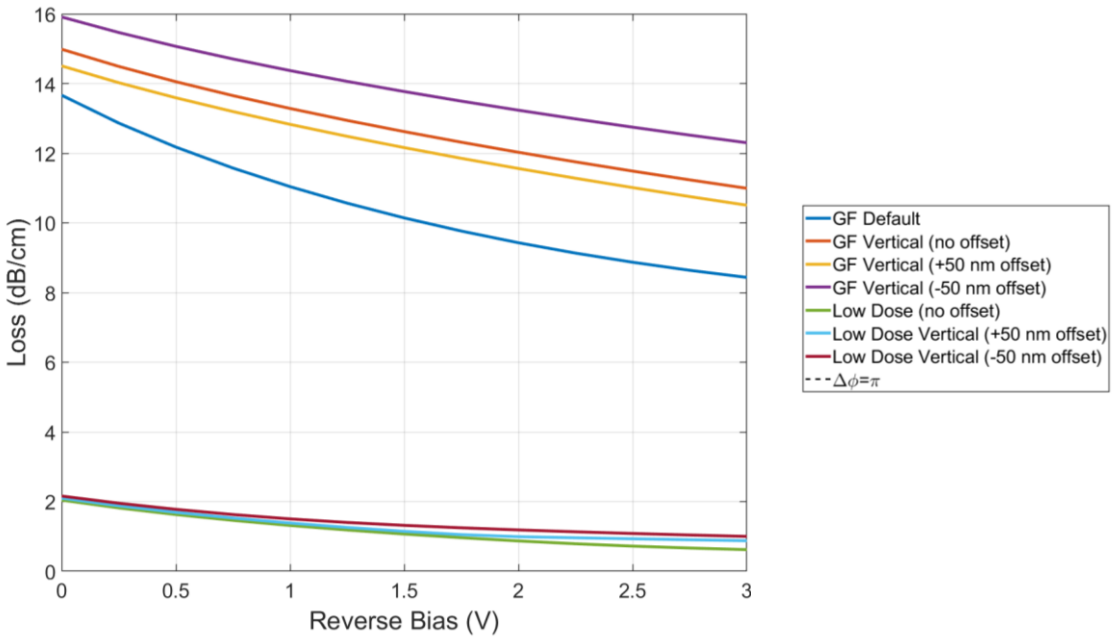


Figure 4.15: Propagation power loss in (dB/cm) as the reverse voltage applied is changed for the different doping profile cases.

The first case (GF default) offers a lower loss when compared to cases 2 – 4, but at the same time from the previous studies, the first has a higher capacitance; thus, the speed of modulation is lower. Comparing between all the different cases studies the vertical doping profile (case 2) has been chosen to be implemented for the modulator, as it has the highest modulation speed compared to other with low $V_{\pi}L$ of 0.74 V.cm and losses in the range of [15 – 12] dB/cm in the proposed operation region.

4.6 Modulator GDS layout and tape-out

A GDS file is prepared to be used by GlobalFoundries for fabricating the proposed design of the Mach-Zehnder Interferometer (MZI) assisted with Ring Resonator (RR). Several additional simulations needed to be conducted to get all the required parameters for the design. Some of those simulations are the same, which was conducted for the athermal MZI. For example, the bend radius used in the MZI-RR modulator design is the same used in the athermal MZI, which is 15 μm. Another design needed is the design of the coupler, a design of an adiabatic coupler was used, which achieved a splitting ratio of 50:50 of the transmitted power. The adiabatic coupler design used was studied by one of the team members. Several constraints have to be taken into consideration as those constraints were due to the fabrication limitation from GF. Figure 4.16 shows the GDS layout indicating the different simulated parts which are highlighted by a dashed line. Also, it shows the full design of the proposed device with the electrical pads used to connect the voltage to the correct corresponding PN region.

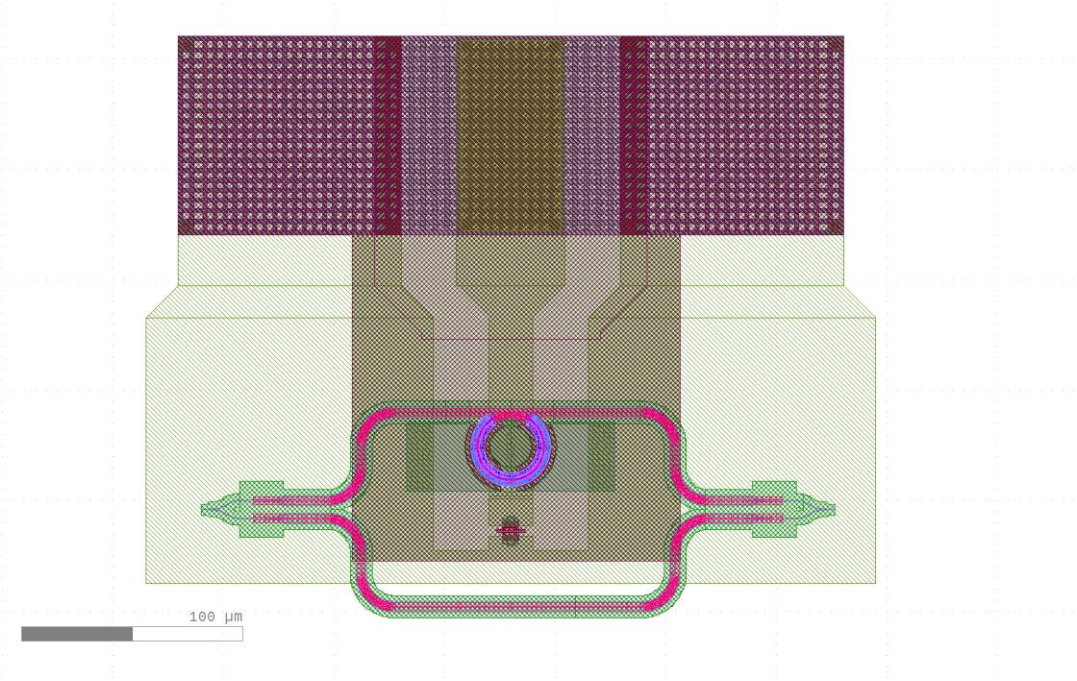
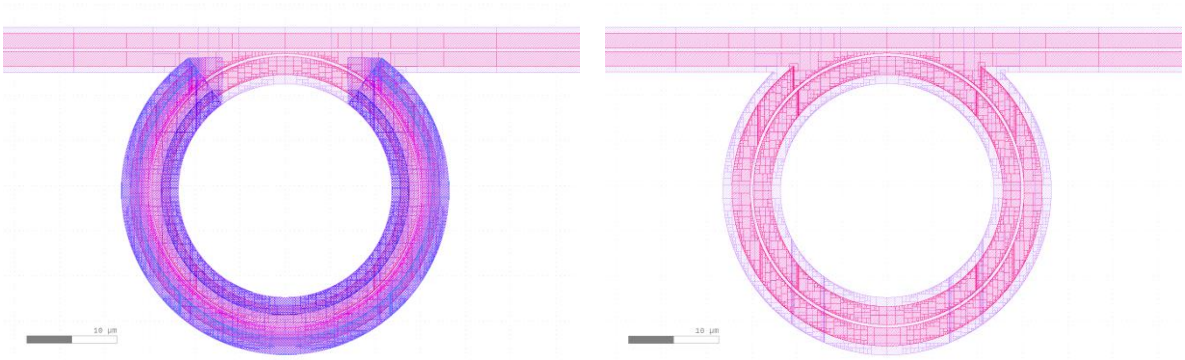


Figure 4.16: GDS file of the proposed MZI-RR modulator with electrical pads

Figure 4.17 shows the Ring resonator coupling light from one of the arms of the MZI with a doping coverage of 75% of the RR. The doping was not applied over the full area of the RR because doping affects the refractive index and the absorption coefficient, as described in section 4.1. Thus, the coupling coefficient will different, and the speed of the modulation will be affected. All of the previous simulations were taking into consideration that the doping region will be covering 75% of the full ring are. The coupling gap between the ring resonator and the MZI is 200 nm; this gap was chosen as it will achieve the highest modulation speed, which is directly linked to the coupling coefficient; hence the coupling gap is fixed. The detailed simulation for the gap calculation is described in section 4.3. Figure 4.18 shows the design of the adiabatic coupler. The proposed design was studied by one of the team members and showed a good splitting ratio of 50:50 for each arm.



(a) *(b)*

Figure 4.17: Ring Resonator design coupled from one the arms of the Mach-Zehnder Interferometer (a) RR with doping regions covering 75% of the RR region (b) RR without doping region.

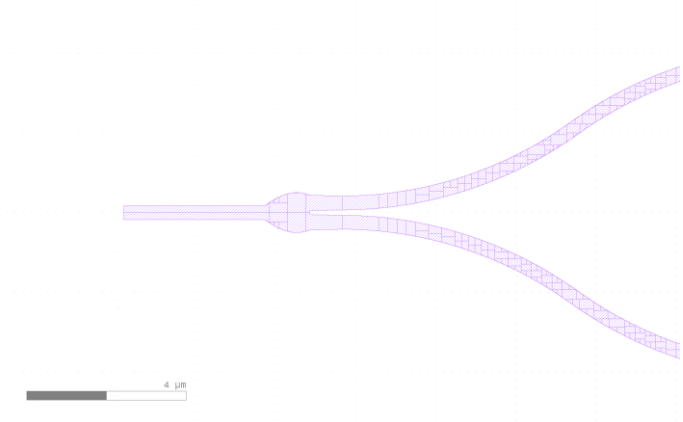


Figure 4.18: Adiabatic coupler for a splitting ratio of 50:50

Chapter 5: Conclusion

5.1 Conclusion

Communication speed in the data center is a critical topic to be addressed as the load on those centers increases. On a chip level, using silicon photonics as a communication method has huge advantages over electrical interconnects. Silicon photonics offer large bandwidth and ultra-high communication speed over the electrical nano wiring. Due to this advantage, among other advantages, silicon photonics mainly modulators are preferred over electrical wiring. At the same time, there are lots of challenges that affect silicon photonics when compared to traditional electrical wiring. Those challenges can be summarized in two main challenges: foot-print size and temperature affect the device behavior.

In this thesis, two design was proposed to address the main challenges. The first proposed design is an athermal Mach-Zehnder Interferometer, which is a thermally stable MZI that can operate in a range of 80 K. The second proposed design is a Mach-Zehnder Modulator assisted by Ring Resonator. This design offers an ultra-high modulation speed of 40 Gbits/s with a small footprint. Both designs were designed and fabricated using GF-45SPCLO and GF-90WG technology platform.

5.1.1 Athermal design using GF-45SPCLO

The first proposed design is an athermal design of MZI using the GF-45SPCLO platform. The footprint area of the design is 130 μm by 1850 μm with an arm length L_1, L_2, L_3 of 1838.9, 1113.5 and 1020.6 in micrometers respectively, and waveguide widths w_1, w_2, w_3 of 558, 400 and 203 in nanometers respectively. The proposed design simulated result shows that the design can operate in a temperature range of [300 – 380 K] without affecting the operation. This design offers a passive temperature compensation for the thermo-optical coefficient by carefully choosing the waveguide widths and the arm lengths to compensate for the temperature change. This behavior is due that the electromagnetic wave (light traveling in the MZI) does not face an optical path difference when the temperature change. It can be put simply that if the temperature increase, both arms will compensate in a way that the light will not suffer from any path difference, thus the proposed device can be athermal.

5.1.2 MZM-RR design using GF-90WG

The second proposed design is a Mach-Zehnder interferometer assisted with a ring resonator. This design reduces the footprint of the overall design by modulating the signal using the ring resonator. At the same time, a high modulation speed is achieved by studying the different doping profiles and the effect of each profile on modulation behavior. This design has a low $V_\pi L$ of 0.74 V.cm and losses in the range of [15 – 12] dB/cm, while at the same time offers a cutoff frequency is about 40.1 GHz at zero volts. The design overcomes the challenge of the size (footprint area) while achieving a high modulation speed.

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